

Dimer-monomer transition defines a novel hyper-thermostable peptidoglycan hydrolase mined from bacterial proteome

Reviewed Preprint

v1 • July 16, 2024

Not revised

Li Zhang, Fen Hu, Zirong Zhao, Xinfeng Li, Mingyue Zhong, Jiajun He, Fangfang Yao, Xiaomei Zhang, Yuxuan Mao, Hongping Wei, Jin He , Hang Yang 

National Key Laboratory of Agricultural Microbiology & Hubei Hongshan Laboratory, College of Life Science and Technology, Huazhong Agricultural University, Wuhan 430070, China • Key Laboratory of Gastrointestinal Cancer (Fujian Medical University), Ministry of Education, Department of Etiology, School of Basic Medical Sciences, Fujian Medical University, Fuzhou 350002, China • Key Laboratory of Virology and Biosafety, Wuhan Institute of Virology, Chinese Academy of Sciences, Wuhan 430071, China • The State Key Laboratory Breeding Base of Basic Science of Stomatology (Hubei-MOST) & Key Laboratory of Oral Biomedicine, Ministry of Education, School of Stomatology, Wuhan University, Wuhan 430079, China • University of Chinese Academy of Sciences, Beijing 100049, China • Hubei Jiangxia Laboratory, Wuhan 430200, China

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

Phage-derived peptidoglycan hydrolases (i.e., lysins) are considered a promising alternative to traditional antibiotics due to their low risks of resistance and unique mechanisms of action. However, the discovery of these enzymes is often hampered by limited source of available phage genomes. Herein, we report a new strategy to mine novel peptidoglycan hydrolases from bacterial proteomes by lysin-derived antimicrobial peptide-primed screening. As a proof-of-concept, five novel peptidoglycan hydrolases from the *Acinetobacter baumannii* proteome (PHAb7-PHAb11) were identified using PlyF307 lysin-derived peptide as a template. PHAb10 and PHAb11 showed potent bactericidal activity against a variety of pathogens even after treatment at 100°C for 1 hour, while the other three were thermosensitive. We solved the crystal structures of PHAb8, PHAb10, and PHAb11 and unveiled that hyper-thermostable PHAb10 underwent a unique folding-refolding thermodynamic scheme mediated by the dimer-monomer transition, while thermosensitive PHAb8 formed a monomer. Two mouse models of bacterial infection further demonstrated the safety and efficacy of PHAb10. Altogether, our antimicrobial peptide-primed strategy provides new clues for the discovery of novel antimicrobial drugs with therapeutic promise.

eLife assessment

This **valuable** study explores a new strategy of lysin-derived antimicrobial peptide-primed screening to find peptidoglycan hydrolases from bacterial proteomes. Using this strategy, the authors identified five peptidoglycan hydrolases from *Acinetobacter baumannii*, which they tested on various Gram-positive and Gram-negative pathogens for antimicrobial activity. The data presented are **solid** and will be of interest to microbiologists.

<https://doi.org/10.7554/eLife.98266.1.sa2>

Introduction

Despite decades of concerted action from all over the world, antimicrobial resistance remains a growing global threat nowadays (1). The unreasonable use of antimicrobial agents in human, livestock, industrial agriculture, and aquaculture is the main reason for the uncontrollable spread and accumulation of antimicrobial resistance (2). Abuse and overuse of antimicrobial agents leads to the accumulation of these antimicrobials in environment, eventually increasing the chances of evolution of bacterial pathogens to acquire resistance (3, 4). Therefore, the concept of “One Health” has been proposed to address the alarming levels of antimicrobial resistance (5, 6). In recent decades, the number of bacteria resistant to first-line drugs has been increasing, and the number of bacteria resistant to second- or third-line drugs has also increased. Some of them have even been identified as multi-drug resistant, extensively drug resistant, or pan-drug resistant isolates (7). For example, methicillin-resistant *Staphylococcus aureus*, vancomycin-resistant *S. aureus*, vancomycin-resistant enterococcus, colistin-resistant Enterobacteriaceae, third generation of cephalosporin-resistant *Escherichia coli* and *Klebsiella pneumoniae*, carbapenem-resistant *E. coli*, *K. pneumoniae*, and *Pseudomonas aeruginosa*, and multi-drug *Acinetobacter baumannii* cause a severe threat and are responsible for most of the morbidity and mortality in modern clinical practice (8). Certainly, the global medical need for novel antimicrobial has yet to be fully addressed.

Recently, many attempts have been made to develop novel antimicrobial drugs with different targets or mechanisms of action than traditional antibiotic (9). These antibacterial pipelines include small molecules with new targets such as ribosomal binding sites, membrane and cell wall, gene interference, and metabolism (10, 11); antibacterial vaccines, antibodies and antibody-drug conjugates targeting conserved antigens of specific pathogen (12); phages and phage-derived peptidoglycan hydrolases, i.e., lysins that can lyse susceptible bacteria individually or as a cocktail (13, 14); microbiome-modulating therapies targeting bacterial colonization environments (primarily the intestine) (15); anti-virulent agents that target different bacterial virulence factors, such as quorum sensing, biofilm formation, adhesion, multiple regulators and persisters (16); potentiators that alone have no or low antibacterial activity but otherwise restore or enhance the activity of another antibacterial agent (17); immunomodulators that modulate the outcome of bacterial infection via a host-directed approaches (18, 19); and nanobiotics that have antibacterial capacity through the generation of reactive oxygen species, cell membrane permeation, triggering DNA damage or interrupting transmembrane electron transport (20). In addition, drug repurposing, the strategy of repurposing non-antimicrobial drugs to treat bacterial infections, also represents a time-saving and cost-effective approach to antimicrobial discovery (21, 22). While these approaches hold particular promise and

advantages in accelerating discovery of potential compounds for clinical studies, they nevertheless have substantial obstacles to overcome. The common denominator is the lack of efficient hit-to-lead discovery methods.

Recent advances in metagenomics, artificial intelligence and synthetic biology have allowed and inspired a renaissance in research into novel antimicrobial agent discovery. There is a trend towards mining encrypted antimicrobial agents, especially peptides, through algorithms coupled with biophysical feature selection that rely on rational design, machine learning, or artificial intelligence-based tools derived from big data, such as the human proteome and intestinal microbiome (23 [↗](#), 24 [↗](#)), human and environmental metagenomics (25 [↗](#)), protein sequence space (26 [↗](#)), and virtual prediction library and display (27 [↗](#), 28 [↗](#)). It is not fully known yet how this event will affect the evolutionary arms race between bacteria and the agents designed to kill them, and consequently the entire field of antimicrobial resistance. But it is clear that novel antimicrobial agents with relatively low risk of resistance will be crucial.

Abundant evidence has showed that bacterial peptidoglycans are a promising target for the development of antimicrobial agents with low risk of inducing resistance. A well-documented example in this regard is the peptidoglycan hydrolases, also known as lysins, derived from phages. In recent years, lysins have been shown to be efficient alternatives to traditional antibiotics for infections caused by Gram-positive bacteria in various animal models (14 [↗](#), 29 [↗](#)). Encouragingly, several lysins targeting *S. aureus* has been evaluated in clinical trials (30 [↗](#)). However, the discovery of lysins against Gram-negative pathogens meeting the needs described in the WHO list of priority pathogens is still in its infancy (31 [↗](#), 32 [↗](#)). Additionally, the discovery of phage-derived peptidoglycan hydrolases has also been hampered by limited sources of published phage genome data. Although not yet tested extensively, recent progress in lysins targeting on Gram-negative bacteria have demonstrated a clue linking antibacterial activity to their internal antimicrobial peptides, by which lysins cross the bacterial outer membrane and contact its underlying peptidoglycan substrates (33 [↗](#)-35 [↗](#)). These observations led us to consider whether it is feasible to mine novel antibacterial peptidoglycan hydrolases or their homologues from bacterial proteomes by using lysin-derived antimicrobial peptide as template. As a proof-of-concept, here we searched for novel peptidoglycan hydrolases from the *A. baumannii* proteome database (PHAbs, Fig. S1 [↗](#)) using the antimicrobial peptide P307 derived from PlyF307 lysin (33 [↗](#)) as a template. Two of them (PHAb10 and PHAb11) were found to be hyperthermally stable even after treatment at 100°C for 1 hour, which might be mediated by a unique monomer-dimer swapping. Furthermore, both were active against a wide variety of clinically relevant bacteria *in vitro* and in mouse infection models. This observation supports our hypothesis that antimicrobial peptide-primed mining strategy was feasible for discovering novel peptidoglycan hydrolases from bacterial proteome with high bactericidal activity and ideal physicochemical properties.

Results

Screening of putative peptidoglycan hydrolases from the *A. baumannii* proteome based on a lysin-derived antimicrobial peptide-primed mining strategy

Since a variety of lysins active against Gram-negative bacteria are shown to harbor an N- or C-terminal antimicrobial peptide, this led us to speculate that antimicrobial peptide would be used as template to mine novel peptidoglycan hydrolases. To prove this hypothesis, P307, a well-documented antimicrobial peptide against *A. baumannii* from PlyF307 lysin (33 [↗](#), 34 [↗](#)), was used as the template to mine putative peptidoglycan hydrolases from *A. baumannii* (PHAb) proteome database publicly available in NCBI (Fig. 1a [↗](#)). Finally, 204 hits were identified and further divided into 5 clades base on phylogenetic analysis. Clade I contained 162 hits with high similarity

(> 85%) to the AcLys lysin from *A. baumannii* 5057UW prophage (Fig. 1b), and clade II-V contained 42 hits with relatively lower homology (40-77%) to each other (Fig. 1b and Fig. S2a).

Due to the high similarity of amino acid sequences within each clade, we selected a putative peptidoglycan hydrolase from each clade for further characterization by analyzing the selectivity priority in evolution and the overall physicochemical properties (water solubility, hydrophilicity, hydrophobicity, charge, flexibility, and rigidity) of each sequence *in silico*. Finally, WP_065188953.1 from clade I, WP_0383495441.1 from clade II, KCY28522.1 from clade III, WP_000109885.1 from clade IV, and WP_032023242.1 from clade V were shortlisted and renamed as PHAb11, PHAb10, PHAb9, PHAb8, and PHAb7, respectively, for further investigation (Fig. 1b). Further homology analysis revealed that these five peptidoglycan hydrolases showed great diversity with the structured AcLys lysin and the original PlyF307 lysin (Fig. S2b).

PHAb10 and PHAb11 are highly thermostable peptidoglycan hydrolases with a broad-spectrum of action

To test the activity of these PHABs, their coding sequences were chemically synthesized, cloned to vector, and expressed in *E. coli* cells. As expected, all five PHABs were well expressed as soluble proteins in *E. coli* BL21(DE3) with high purity (Fig. S3a). Next, we examined the hydrolytic activity of these 5 PHABs on peptidoglycan derived from *A. baumannii* 3437 by halo assay. Results showed that even at a low concentration of 2.5 µg/ml, all PHABs formed clear zones on autoclaved *A. baumannii* lawns (Fig. S3b), indicating that these PHABs were active against peptidoglycan from *A. baumannii*. To further confirm the ability of these PHABs against peptidoglycan from different bacteria, we measured the area of clear zones generated by different concentrations of PHABs on each lawn of two isolates of *A. baumannii*, *P. aeruginosa*, and *E. coli*. Encouragingly, these PHABs were active against peptidoglycans from all strains tested and formed a clear halo (Fig. 2a). Among these PHABs, PHAb10 and PHAb11 showed larger size of clear zones under each concentration tested (Fig. 2a), indicating that these two PHABs might have stronger peptidoglycan hydrolyzing activity. To prove this, we further evaluated the bactericidal activity of these PHABs against each isolate of *A. baumannii*, *P. aeruginosa*, and *E. coli* by a log-killing assay. Consistent with the halo assay, PHAb10 and PHAb11 exhibited outstanding bactericidal activity against all strains tested, with a 3.5-4.6 log reduction in viable bacterial counts after treatment for 1 hour with 50 µg/ml of each enzyme (Fig. 2b). Like other lysins, the potent bactericidal activity of PHAb10 and PHAb11 was time-dependent. After treatment with 50 µg/ml of either enzyme, a ~0.9 log reduction in viable *A. baumannii* 3437 for both enzymes were observed within the first 30 seconds, with a substantial reduction to 3.1 logs (from 6.0 log to 2.9 log) for PHAb10 and 2.5 logs (from 6.0 log to 3.5 log) for PHAb11 after 15 minutes (Fig. 2c).

Since previous reports documented the diverse performance of lysins against Gram-negative and Gram-positive bacterial species in exponential and stationary phases (33), we therefore examined the bactericidal activity of PHAb10 and PHAb11 against different bacteria in different growth phases. Interestingly, robust bactericidal activity was observed for both enzymes in exponential and stationary phase cultures of *A. baumannii*, *P. aeruginosa*, *E. coli*, *K. pneumoniae*, *S. suis*, and *E. faecalis* (Fig. 2d). It should also be noted that exponential bacteria were more susceptible to these two enzymes than stationary bacteria, similar to observations reported elsewhere (33). However, for *S. aureus*, detectable susceptibility to PHAb11 was only observed in its exponential phase (Fig. 2d). In addition, PHAb10 and PHAb11 exhibited dose-dependent killing against sensitive Gram-negative and Gram-positive strains tested. Specifically, treatment with 6.25 µg/ml PHAb10 for 1 hour resulted in a ~3 log reduction in *A. baumannii*, a ~2.7 log reduction in *P. aeruginosa*, a ~3 log reduction in *E. coli*, ~1.08 log reduction in *K. pneumoniae*, and a ~0.94 log reduction in *S. suis*, and treatment with 6.25 µg/ml PHAb11 for 1 hour caused a ~2.8 log reduction in *A. baumannii*, a ~2.6 log reduction in *P. aeruginosa*, a ~2.7 log reduction in *E. coli*, a ~0.69 log reduction in *K. pneumoniae*, a ~1.3 log reduction in *S. aureus*, and a ~2.15 log

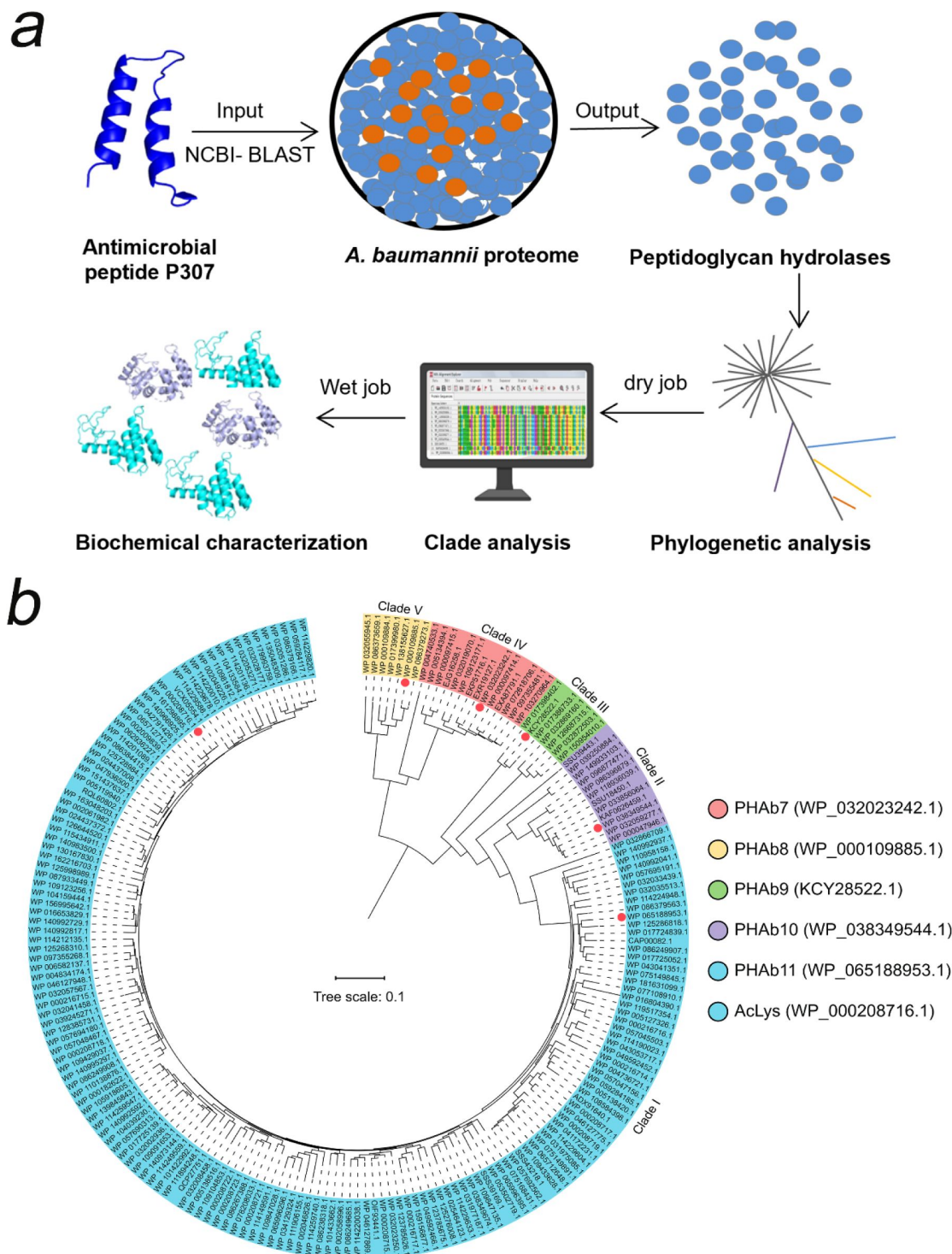


Fig. 1

Screening of putative peptidoglycan hydrolases from the *A. baumannii* proteome.

(a) Workflow for the development of an integrated system for peptidoglycan hydrolase screening. (b) Phylogenetic analysis of putative peptidoglycan hydrolases from the *A. baumannii* proteome database. Multiple sequence alignments are performed using MEGAX software with ClustalW algorithm, and the Neighbor-joining method is used to construct a phylogenetic tree. Classified clades are labeled in different colors and representative peptidoglycan hydrolases from each clade are pointed by colored circles.

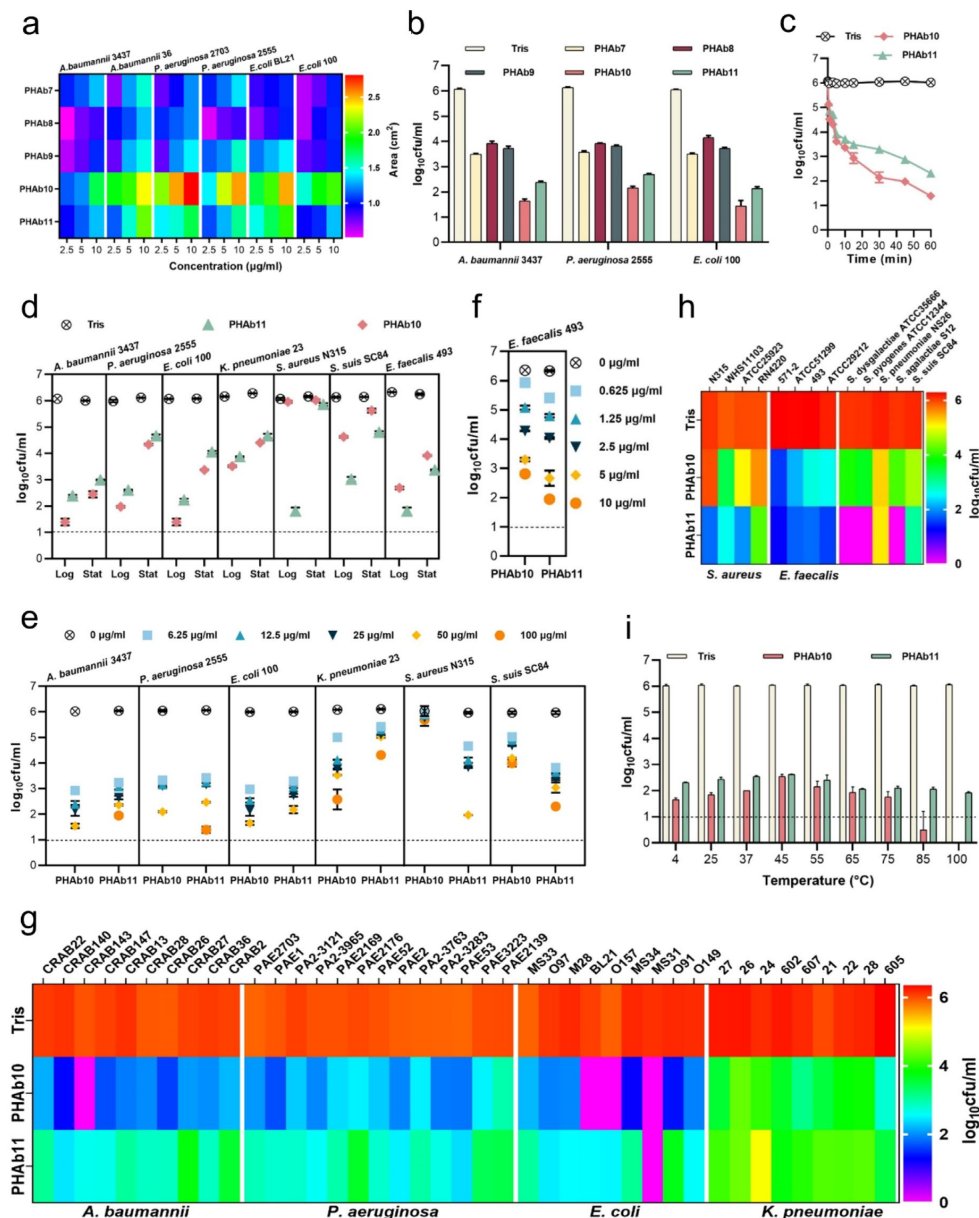


Fig. 2

PHAb10 and PHAb11 are highly thermostable peptidoglycan hydrolases with a broad-spectrum of action.

(a) Heat map showing the halo size formed by different concentrations of peptidoglycan hydrolases on different bacterial lawns. (b) Bactericidal activity of five peptidoglycan hydrolases (50 µg/ml) in 20 mM Tris-HCl (pH 7.4) at 37°C for 1 hour against different exponential bacteria. (c) Time-dependent bactericidal activity of 50 µg/ml PHAb10 or PHAb11 against exponential *A. baumannii* in 20 mM Tris-HCl (pH 7.4). (d) Bactericidal activity of 50 µg/ml PHAb10 or PHAb11 in 20 mM Tris-HCl (pH 7.4) at 37°C for 1 hour against different bacteria in exponential and stationary phases. (e-f) Dose-dependent bactericidal activity of PHAb10 and PHAb11 against multiple exponential bacteria in 20 mM Tris-HCl (pH 7.4) at 37°C for 1 hour. (g-h) Susceptibility of various Gram-negative (g) and Gram-positive (h) bacterial strains to PHAb10 and PHAb11. Exponential cultures of each bacterium are treated with 50 µg/ml PHAb10 or PHAb11 for 1 hour at 37°C and residual viable bacterial cells are counted by plating serial dilutions onto agar plates. For *E. faecalis*, 10 µg/ml of each peptidoglycan hydrolase is used. (i) Thermal stability of PHAb10 and PHAb11. Each enzyme is stored at different temperatures for 1 hour, cooled to room temperature, and then incubated with exponential *A. baumannii* 3437 in 20 mM Tris-HCl (pH 7.4) at a final concentration of 50 µg/ml at 37°C for 1 hour. Viable bacteria are counted after each treatment by plating serial dilutions on LB agar. Dash lines represent the limit of detection and data below the limit of detection is not shown.

reduction in *S. suis* (**Fig. 2e**). Interestingly, *E. faecalis* was more sensitive to PHAb10 and PHAb11, showing a ~ 3.8 log and a ~ 4.6 log reduction respectively after treatment with 10 $\mu\text{g/ml}$ of each enzyme for 1 hour (**Fig. 2f**).

To further confirm the bactericidal activity of PHAb10 and PHAb11, the susceptibility of various clinical isolates such as *A. baumannii*, *P. aeruginosa*, *E. coli*, *K. pneumonia*, *S. aureus*, and *E. faecalis*, as well as representative isolates of *S. dysgalactiae*, *S. agalactiae*, *S. pyogenes*, *S. pneumoniae*, and *S. suis* to both enzymes were examined by log-killing assay. Our results showed that all tested strains, whether antibiotic-resistant or not, were sensitive to PHAb10 and PHAb11 (**Fig. 2g-h**). Importantly, Gram-negative bacteria were more sensitive to PHAb10, with 3.9-6.0 log reductions for *A. baumannii* isolates, 3.7-5.0 log reductions for *E. coli* isolates, 2.4-4.3 log reductions for *P. aeruginosa* isolates, and 1.1-3.6 log reductions for *K. pneumoniae* isolates (**Fig. 2g**). In contrast, PHAb11 was more bactericidal than PHAb10 against Gram-positive bacteria, causing 1.8-4.3 log reductions for *S. aureus* isolates, 4.5-5.0 log reductions for *E. faecalis* isolates, and 1.0-6.1 log reductions for various streptococcal strains (**Fig. 2h**). Further biochemical characterization showed that PHAb10 and PHAb11 remained highly active at pH 5.0-10.0, and the maximum bactericidal activity was observed at pH 7.0 for both enzymes (**Fig. S3c**). Similar to other previously reported lysins (36, 37), PHAb10 and PHAb11 were sensitive to NaCl and urea in a dose-dependent inhibitory manner (**Fig. S3d**). Surprisingly, both enzymes retained outstanding bactericidal activity after heat treatment at temperature up to 100°C for 1 hour (**Fig. 2i**). Indeed, after 1 hour treatment at 100°C, the bactericidal activity was improved slightly for both enzymes. In contrast, PHAb7, PHAb8, and PHAb9 completely lost their activity after treatment at 70°C for 1 hour (**Fig. S3e**). Together, these observations showed that PHAb10 and PHAb11 were highly thermostable peptidoglycan hydrolases with a broad spectrum of action.

Different action mechanisms of PHAb10 and PHAb11 in killing Gram-negative and Gram-positive bacteria

Due to the natural barrier of the outer membrane, several peptidoglycan hydrolases (i.e., lysins) have been reported to have no or low bactericidal activity against Gram-negative bacteria in the absence of outer membrane penetrating agents (31, 32, 38). However, for most of these lysins, their activity is almost exclusively restricted to Gram-negative bacteria (39, 40), which casts doubt on the mechanisms of action of PHAb10 and PHAb11 in killing both Gram-negative and Gram-positive bacteria. Structural prediction analysis showed that PHAb10 contained a putative lysozyme catalytic domain (amino acids 1-110, P10-Lys) and a C-terminal cationic peptide (amino acids 111-149, P10-CP), whilst PHAb11 contained an N-terminal cationic peptide with unknown function (amino acids 1-36, P11-NP), followed by a putative lysozyme catalytic domain (amino acids 37-145, P11-Lys) and a C-terminal cationic peptide (amino acids 146-184, P11-CP) (**Fig. 3a**). Studies have shown that the cationic region of phage lysins can facilitate their penetration into the outer membrane, thereby fulfilling their bactericidal activity against Gram-negative bacteria (33, 35, 37). To examine whether it is the reason for the killing ability of PHAb10 and PHAb11 against Gram-negative bacteria, we further analyzed *in silico* the physicochemical properties of these putative cationic peptides harbored by these two enzymes and found that all three peptides, that is, P10-CP, P11-CP, and P11-NP might act as antimicrobial peptides (**Table S1**). Therefore, we synthesized these peptides and examined their antibacterial activity by log-killing assay. As shown in **Fig. 3b**, we observed bactericidal activity in P10-CP and P11-CP against *A. baumannii*, *P. aeruginosa*, *S. aureus*, and *E. faecalis* in both the exponential and stationary phases. However, rare lytic activity was observed in P11-NP, suggesting that a potential role for it in functions other than bactericidal activity. Notably, P11-CP was more robust than P10-CP (**Fig. 3b**), which might be due to an extra positively charged amino acid (i.e., Lysine) in P11-CP (**Table S1**). More importantly, the putative lysozyme catalytic domains of two enzymes, P10-Lys, P11-Lys, and P11-N showed rare bactericidal activity against all Gram-negative and Gram-positive bacteria tested (**Fig. 3c-d**), which indicated that the antimicrobial peptides, namely P10-CP and P11-CP, were essential for the bactericidal activity of both enzymes. Curiously, for

unknown reasons, P11-C showed higher bactericidal activity than the holoenzyme PHAb11 against all bacteria tested (Fig. 3d). Considering the different targets of antimicrobial peptide and peptidoglycan hydrolase, we further asked the killing mechanism of susceptible bacteria by PHAb10 and PHAb11. To this end, we evaluated the effects of different truncations of PHAb10 and PHAb11 on different bacteria by halo assay to test their peptidoglycan hydrolase activity. Results showed that lysozyme-containing truncations from both enzymes showed hydrolytic activity only towards peptidoglycan from Gram-negative bacteria but not from Gram-positive bacteria (Fig. 3e), suggesting that the killing of Gram-positive bacteria by PHAb10 and PHAb11 is probably mediated mainly by their internal antimicrobial peptides, i.e., P10-CP and P11-CP. Supporting to this assumption, rare peptidoglycan hydrolytic activity was observed in P10-CP and P11-CP (Fig. 3e). Altogether, these results suggested that the killing of Gram-negative bacteria by PHAb10 and PHAb11 might be a combination of lysozyme-mediated peptidoglycan hydrolytic activity and antimicrobial peptide-mediated outer membrane-disruption activity. However, their bactericidal activity against Gram-positive bacteria was primed solely by their internal antimicrobial peptides (Fig. 3f).

PHAb8 is a thermosensitive monomer, whilst PHAb10 and PHAb11 formed a thermostable dimer

To understand the catalytic mechanism and thermal stability of these novel peptidoglycan hydrolases, we solved the crystal structures of PHAb8, PHAb10, and PHAb11 by X-ray crystallography (Table S2), and the results showed that PHAb8 was a monomer (Fig. 4a and Fig. S4a), whilst PHAb10 (Fig. 4b) and PHAb11 (Fig. 4c) formed an asymmetric dimer. Specifically, the two asymmetric units of PHAb10 formed an antiparallel dimer with a rotation angle of 177.2° along the rotation axis and a distance of 10.5 Å between them. Further, PHAb10 dimer adopted a circularly permuted architecture, in which the N-terminus of one chain and the C-terminus of another chain formed a cavity, which might be the binding site of the peptidoglycan substrate (Fig. 4b). The two subunits of PHAb11 formed a tail-to-tail dimer with a rotation angle of -179.4° along the rotation axis and a distance of 37.1 Å between them. Each PHAb11 monomer contained an independent substrate-binding cavity around its catalytic triad (Fig. 4c). Notably, the N-terminal 36 amino acids, P11-NP, were missing in our PHAb11 dimer. We speculated that it may be destroyed during crystal formation. Interestingly, PHAb8 showed high topological similarity to monomeric PHAb10 and PHAb11, with the typical catalytic triad (Glu-Asp-Thr) located at the coil region (Fig. 4d and Fig. S4a-c). Dali-based screening further showed that the catalytic triad of PHAb8, PHAb10, and PHAb11 were highly conserved in homologous proteins, including AcLys (PDB ID 6ET6), LysF1 (PDB ID 7M5I), SpmX-Mur-Ae (PDB ID 6H9D), P22 (PDB ID 2ANX), and R21 (PDB ID 3HDE) (Fig. S4d and Table S3). Superimposition of PHAb8, PHAb10, PHAb11 with other well-known lysozymes, including hen egg-white lysozyme (HEWL, PDB ID 4HPI), goose egg-white lysozyme (GEWL, PDB ID 153L), T4 lysozyme (T4L, PDB ID 1LYD), and lambda lytic transglycosylase (PDB ID 1LYD) revealed that the overall structures of PHAb8, PHAb10, and PHAb11 had relatively low similarity to HEWL, GEWL, and lambda lytic transglycosylase (Fig. S4e-g). Despite their lower amino acid sequence similarity (Fig. S4h), PHAb8, PHAb10, and PHAb11 superimposed well with T4 lysozyme, especially in their Glu-Asp-Thr catalytic triad regions (Fig. S4i and Table S3), suggesting that PHAb8, PHAb10, and PHAb11 may be phage-originated. Supporting to this, all three enzymes were predicted to be originated from prophages by PhageBoost (Fig. S5). All these observations together indicated that PHAb8, PHAb10, and PHAb11 were highly evolutionarily related to T4 lysozyme and might share similar catalytic mechanisms of action.

To reveal which domain contributes to the thermal stability of PHAb10 and PHAb11, we examined the residual bactericidal activity of their individual domains by treating them at 100°C for 1 hour. Results showed that for all strains tested, the holoenzyme and its antimicrobial peptide domain P10-CP retained high bactericidal activity, but its lysozyme catalytic domain P10-Lys did not (Fig.

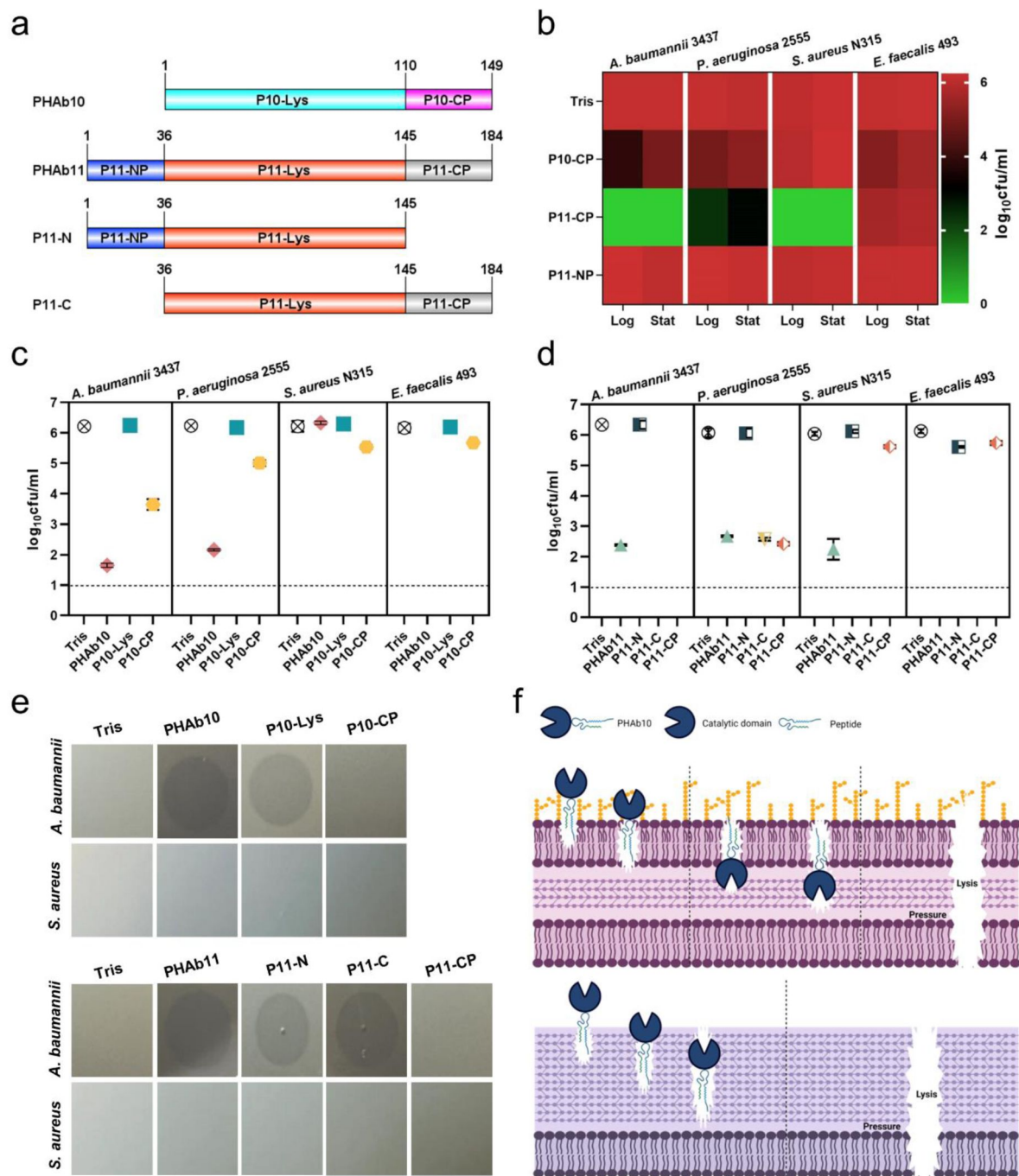


Fig. 3

Mechanism of action of PHAb10 and PHAb11 on Gram-negative and Gram-positive bacteria.

(a) Schematic representation of PHAb10 and PHAb11 domains. (b) Bactericidal activity of peptides derived from PHAb10 and PHAb11. Bacterial cells are treated with 50 µg/ml P10-CP, P11-NP, or P11-CP in 20 mM Tris-HCl at 37°C for 1 hour. (c-d) Bactericidal activity of different truncations of PHAb10 and PHAb11. Exponential *A. baumannii* 3437 cells are incubated with 50 µg/ml of each truncated fragment in 20 mM Tris-HCl for 1 hour at 37°C. Dash lines represent the limit of detection and data below the limit of detection is not shown. (e) Peptidoglycan hydrolytic activity of different truncations of PHAb10 and PHAb11. 0.1 µg of each truncated fragment is dropped onto autoclaved lawns of *A. baumannii* 3437 and *S. aureus* N315 and incubated at 37°C for 4 h. The groups treated with an equal volume of 20 mM Tris-HCl served as controls. (f) Schematic bactericidal mechanisms of PHAb10 against Gram-negative and Gram-positive bacteria.

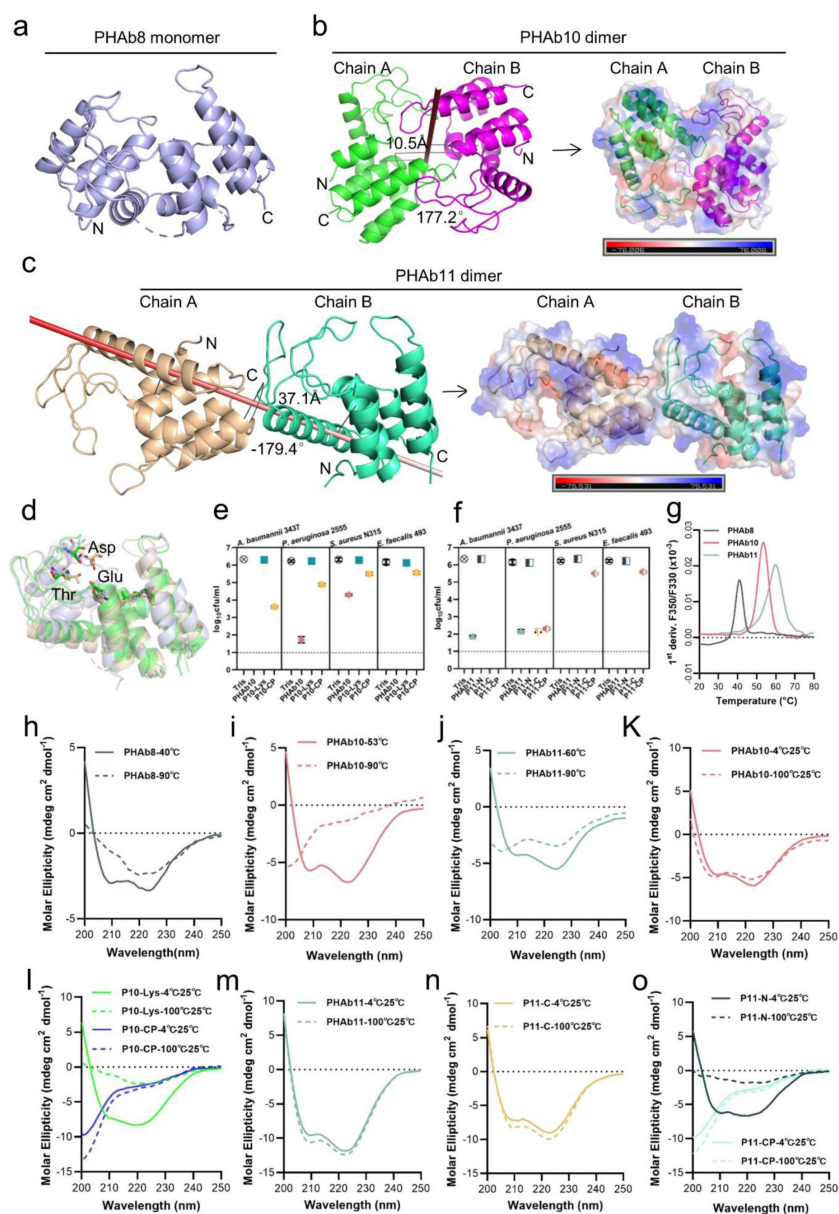


Fig. 4

PHAb10 and PHAb11 are thermostable dimers.

(a) Overall structure of PHAb8 monomer. (b) Dimeric structure of PHAb10. Chain A is colored in green, Chain B is colored in magenta, and the gray stick denotes the rotation axis. The electrostatic surface of PHAb10 with two potential substrate binding cavities are shown (negative = red; positive = blue). (c) Dimeric structure of PHAb11. Chain A is colored in wheat, Chain B is colored in green-cyan, and the red stick denotes the rotation axis. The electrostatic surface of PHAb11 with two potential substrate binding cavities is also shown (negative = red; positive = blue). (d) Superimposition of monomeric PHAb8, PHAb10, and PHAb11. PHAb8 is shown in light blue, PHAb10 in green, and PHAb11 in wheat. Residues involved in the catalytic triad are shown in sticks. (e-f) Residual bactericidal activity of PHAb10, PHAb11, and their truncation variants. Each truncation variant is treated at 100°C for 1 hour, stored at 25°C for 1 hour, and then tested for bactericidal activity by a log-killing assay. Dash lines represent the limit of detection and data below the limit of detection is not shown. (g) Thermal unfolding curves of PHAb8, PHAb10, and PHAb11 from 20 to 100°C as determined by nanoDSF. Values on the y-axis represent the first derivative of the fluorescence ratio at 350 nm and 330 nm. Peaks represent the transition temperature of each protein. (h-j) Circular dichroism spectra of PHAb8 (h), PHAb10 (i), and PHAb11 (j) at 90°C and temperatures close to their transition temperatures. (k-o) Circular dichroism spectra of PHAb10, PHAb11, and their truncation variants before and after heat treatment. Each domain fragment is treated at 100°C for 1 hour, stored at 25°C for additional hour prior to CD detection.

4e). Except for the N-terminal peptide P11-NP, other truncated PHAb11 still maintained substantial bactericidal activity after heating at 100°C for 1 hour (Fig. 4f). These observations raised the question of whether PHAb10 and PHAb11 dimers are hyper-thermostable. To test this, we determined the temperature-dependent denaturation of these enzymes by nano-differential scanning fluorimetry (nanoDSF). Our results showed that PHAb8, PHAb10, and PHAb11 were denatured with transition temperatures of 41.4, 53.7, and 59.9°C, respectively (Fig. 4g). Interestingly, signs of aggregation were only observed in PHAb8 but not in PHAb10 and PHAb11 (Fig. 54j), indicating that PHAb10 and PHAb11 underwent structural refolding without precipitation during heat treatment. Consistent with these observations, the corresponding circular dichroism (CD) curves of PHAb8, PHAb10, and PHAb11 at 90°C were significantly different from their corresponding curves at temperature close to their transition temperature (Fig. 4h-j), suggesting that, similar to PHAb8, PHAb10 and PHAb11 lost their native conformation at high temperature such as 90°C. To further understand the thermodynamics of PHAb10 and PHAb11, we examined the structure of PHAb10 before and after heat treatment at 100°C for 1 hour by CD. Intriguingly, only minor differences were observed before and after heat treatment (Fig. 4k), indicating that PHAb10 almost refolded to its native conformation during cooling after heat treatment. The C-terminal antimicrobial peptide P10-CP of PHAb10 could withstand heat treatment, but P10-Lys was deformed after being treated at 100°C for 1 hour (Fig. 4l). Similar to the folding-refolding phenomenon of PHAb10, PHAb11 and its N-terminal peptide-deleted variant P11-C could restore to their native conformation after 1 hour treatment at 100°C (Fig. 4m-n), whilst its C-terminal peptide-deleted variant P11-N could not refold to its native conformation (Fig. 4o), indicating that C-terminal peptide was essential for its hyper-thermostability. Altogether, these observations suggested that dimerized PHAb10 and PHAb11 possessed unique folding-refolding thermodynamic mechanism.

Folding-refolding thermodynamic of PHAb10 dimer is governed by intermolecular interactions

Since PHAb10 dimer and PHAb11 dimer exhibit a similar folding-refolding thermodynamic profiles, we wondered whether it was achieved through dimer-monomer switching. Therefore, we used PHAb10 as an example to test this hypothesis. We carefully examined the topology of the PHAb10 dimer and found that it was supported by 7 pairs of intermolecular H-bonds, with 3 pairs supporting the head, each 2 pairs in midbody and tail of the PHAb10 dimer (Table S4). Specifically, in the top view of chain A, Arg20 and Gly27 were involved in intermolecular H-bond interactions, supporting the head region, Gly128 and Gly129-mediated interaction contributed to the stability of midbody, whilst Lys133 and Arg137 were involved in intermolecular H-bond forces anchoring the tails of the PHAb10 dimer (Fig. 5a and Table S4). Next, we constructed several PHAb10 variants involving depletion of intermolecular force at the head, midbody, tail, or entire entity of the PHAb10 dimer (Table S5). Since PHAb10 was a verified dimer after chemical cross-linking in SDS-PAGE gel (Fig. 5b), we further examined the polymerization of these PHAb10 variants by Native-PAGE. As shown in Fig. 5c, wildtype PHAb10 showed two bands corresponding to its monomeric and dimeric forms. Whilst the variants R20IG27D, which has mutations in the head-supporting site, and G27DR137I, which has mutations in both the head and tail supporting sites, lost dimerization ability. As a result, both variants lost most of their bactericidal activity after treatment at 100°C for 1 hour (Fig. 5d), which could be further explained by the failure to recover their native forms after heat treatment (Fig. 5e-f). In support of these observations, the variant PHAb10-mut containing mutations in all 7 pairs of dimer-supporting sites showed aborted bactericidal activity (Fig. 5d) and failed to recover its native conformation (Fig. 5e-f) after treatment at 100°C for 1 hour. In contrast, the variants G128D/G129D with mutations in the midbody-supporting sites still showed a mixture of monomers and dimers (Fig. 5c), indicating that depletion of the midbody-supporting force hardly affected the dimerization of PHAb10. Consistent with this observation, the variant G128D/G129D still remained intact bactericidal activity (Fig. 5d) and could refold to its native form after heat treatment (Fig. 5e-f). Moreover, the variant K133A/R137I with mutations in the tail-supporting

site also showed aborted bactericidal activity (**Fig. 5d**) and an irreversible conformation change after heat treatment (**Fig. 5e-f**), although it maintained a weak dimerization ability (**Fig. 5c**). This suggested that intermolecular forces at the tail of the PHAb10 dimer had only a small impact on its thermodynamics properties. Taken all these observations together, we proposed a model by which the PHAb10 dimer executed its folding-refolding thermodynamics (**Fig. 5g**). At temperatures above its transition point (i.e., >53.7°C), dimerized PHAb10 dissociated into monomers without precipitation. During the cooling process, monomeric PHAb10 refolded to a stable dimer *via* the intermolecular interactions between the two monomers, i.e., three pairs of intermolecular bonds at the head, two in the midbody, and two at the tail of the PHAb10 dimer. The folding and refolding dynamics of PHAb10 after heat treatment were somewhat similar to the opening and closing of zippers, where the intermolecular bonds act as inducers.

PHAb10 efficiently removes *A. baumannii* from infection sites *in vivo*

Since PHAb10 showed rare cytotoxicity on cell level (**Fig. S6**), we further tested its bactericidal activity in two different mouse infection models. In the burn wound model of *A. baumannii* infection (**Fig. 6a**), mice were first treated with 10 µg/mouse PHAb10 (10 µl; n = 5) for 4 hours, and then infected with 10 µl 5×10^8 cfu/ml of *A. baumannii* 3437 for 24 hours. We found significantly less viable bacteria remaining at the site of infection compared to Tris-treated controls (**Fig. 6b**). Treatment with 4 µg/mouse minocycline (10 µl; n = 5) also showed good bacterial removal compared to the control group, however, no statistical difference was observed between groups treated with 10 µg/mouse PHAb10 and 4 µg/mouse minocycline (**Fig. 6b**), suggesting that when administered topically, PHAb10 was as efficient as minocycline in clearing susceptible bacteria from the site of infection. In the mouse abscess model, mice were infected hypodermically (i.h.) with 25 µl of *A. baumannii* 3437 at a concentration of 5×10^8 cfu/ml for 24 hour, and then treated with 10 µg/mouse PHAb10 (10 µl; n = 6) or 4 µg/mouse minocycline (10 µl; n = 7) subcutaneously for 5 days. Six days after infection, viable bacteria were recovered from the infection site and counted (**Fig. 6c**). Our results showed that there was rare difference between the minocycline-treated group and the Tris-treated control group (**Fig. 6d**). In contrast, a significant reduction in residual *A. baumannii* was observed in the PHAb10-treated group (**Fig. 6d**), indicating that PHAb10 had better bactericidal efficacy than minocycline when injected subcutaneously at the site of infection in a skin infection model. Notably, rare differences were observed in changes in animal body weight from groups treated with either PHAb10, minocycline, or Tris buffer (**Fig. 6e**). Taking all these observations together, PHAb10 appeared to exhibit good bacterial clearance efficacy *in vivo* and might represent a novel antimicrobial agent for the treatment of drug-resistant bacteria.

Discussion

The uncontrollable spread and emergence of antimicrobial resistance requires the discovery of novel antimicrobial agents with different targets or mechanisms of action from traditional antibiotics. Policy makers, scientists, enterprisers, and investigators have worked together for decades to exploit the ‘trojan horse’ globally, but new options for treating antimicrobial resistance in the clinic remain to be seen. Bacteriophage-derived peptidoglycan hydrolases (i.e., lysins) have been considered as promising alternatives against drug-resistant bacteria due to their potent bactericidal activity, low risks of drug resistance, and unique mechanism of action, as witnessed by several ongoing clinical trials targeting on Gram-positive bacteria. However, for Gram-negative bacteria, the effects of phage-derived lysins were often hampered by their outer membranes, which requires more strategies to overcome. In the present work, we report a new strategy to mine novel bactericidal peptidoglycan hydrolases from bacterial proteomes through lysin-derived antimicrobial peptide-priming screening. As a proof-of-concept, we successfully discovered two

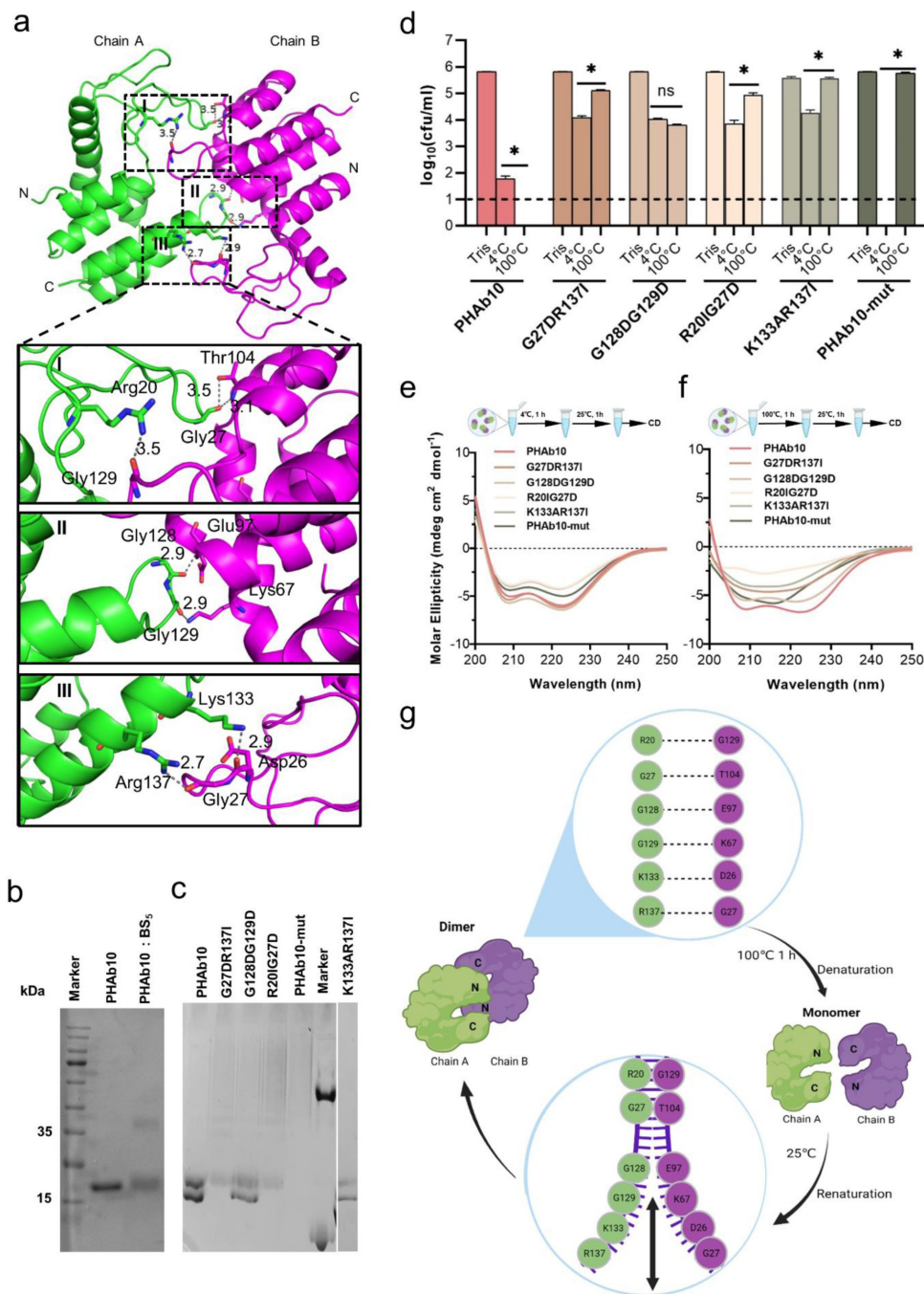


Fig. 5

Folding-refolding thermodynamics of PHAb10.

(a) Intermolecular interactions of PHAb10 dimer. The residues involved are shown as sticks. (b) SDS-PAGE analysis of PHAb10 with or without chemical cross-linker. (c) Native-PAGE analysis of PHAb10 and its variants. Marker: the upper band represents BSA (66.4 kDa), and the lower band represents lysozyme (14 kDa). (d) Bactericidal activity of PHAb10 and its variants before and after heat treatment. Each protein is incubated for 1 hour at 4 or 100°C, stored at 25°C for an additional 1 hour, and then examined for bactericidal activity in 20 mM Tris-HCl (pH 7.4) at a concentration of 50 µg/ml against exponential *A. baumannii* 3437 cells for 1 hour. Dash lines represent the limit of detection and data below the limit of detection is not shown. Data are analyzed by two-tailed Student's t-tests. ns: statistically not significant; *: $p < 0.05$. (e-f) Circular dichroism spectra of PHAb10 and its mutants before and after treatment at 100°C for 1 hour. (g) Zipper model showing the thermodynamics of PHAb10 dimer.

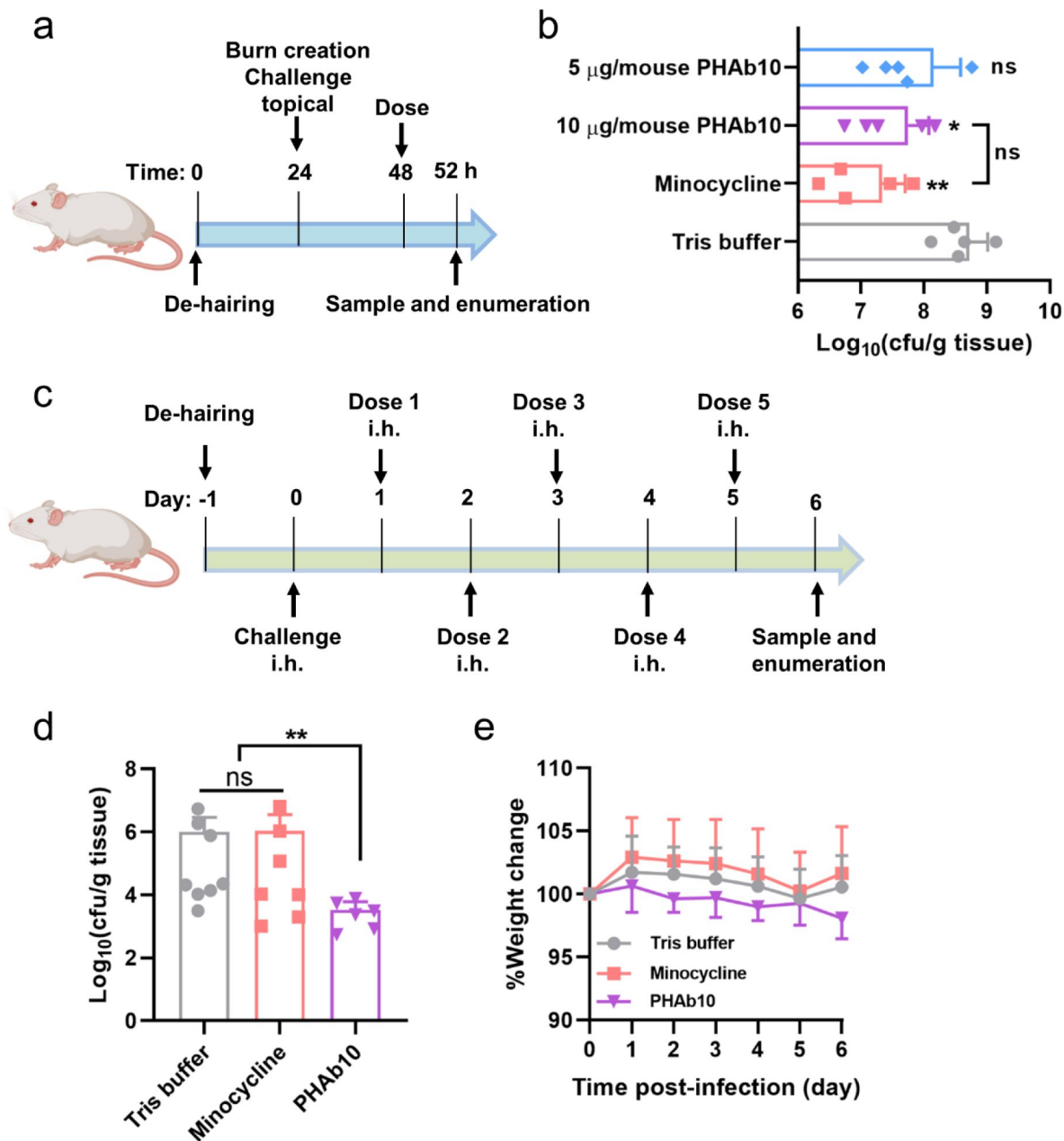


Fig. 6

PHAb10 shows high efficacy in mouse infection models.

(a) Experimental schema for burn wound infection model. Burn wounds produced by exposing naked skin to 65°C water for 12 s was infected with 10 μ l 5×10^8 cfu/ml of *A. baumannii* 3437. At 24-hour post-colonization, mice are treated with 5 or 10 μ g/mouse PHAb10 (10 μ l; n = 5), 4 μ g/mouse minocycline (10 μ l; n = 5), or an equal volume of Tris buffer (10 μ l; n = 5). Four hours post-treatment, viable bacteria on the burn wound skin is collected and counted. (b) Viable *A. baumannii* collected from burn wound skin after each treatment. The number of viable cells in each group is normalized and compared with the number of viable cells treated with Tris buffer by ANOVA. ns: statistically not significant; *: p<0.05; **: p<0.01. (c) Experimental schema for abscess model. Mice are infected hypodermically with 25 μ l 5×10^8 cfu/ml of *A. baumannii* 3437 on the right side of the back dorsum. Twenty-four hours after infection, mice are injected hypodermically with 10 μ g/mouse PHAb10 (10 μ l; n = 6), 4 μ g/mouse minocycline (10 μ l; n = 7), or an equal volume of Tris buffer (10 μ l; n = 8). Each group is injected subcutaneously once a day for 5 consecutive days. Viable bacteria in site of infection are counted 24 hour after the last dose. (d) Viable *A. baumannii* collected from skin after each treatment. The number of viable cells from each group is normalized and compared with the number of viable cells treated by the Tris buffer by ANOVA. ns: statistically not significant; **: p<0.01. (e) Changes in mice body weight per treatment group.

novel peptidoglycan hydrolases from the *A. baumannii* proteome using PlyF307 lysin-derived peptide as a template, highlighting the utility of this strategy in mining antimicrobial agents from bacterial database.

Notably, PHAb10 and PHAb11 exhibited high bactericidal activity against a variety of Gram-negative and Gram-positive microbes. Unlike other Gram-negative lysins that act in an outer membrane permeabilizer-dependent manner (41 [↗](#)-44 [↗](#)), PHAb10 and PHAb11 exhibited intrinsic activity against Gram-negative and Gram-positive pathogens *via* different mechanisms of action. Due to their C-terminal antimicrobial peptides, namely, P10-CP and P11-CP, PHAb10 and PHAb11 were able to penetrate the outer membrane of Gram-negative bacteria in an auto-priming manner, allowing them to access and cleave peptidoglycan bonds, ultimately leading to osmotic lysis of Gram-negative pathogens. For Gram-positive bacteria, they are primarily killed through the action of intrinsic antimicrobial peptides. Interestingly, we noted a difference in the killing of Gram-positive bacteria by PHAb10 and PHAb11, which may be due to the fact that P11-CP had one more basic amino acid than P10-CP, so it had stronger bactericidal activity. In addition, compared to other naturally occurring lysins such as PlyPa03, PlyPa91, and PlyE146 (37 [↗](#), 45 [↗](#)), PHAb10 and PHAb11 showed traceable bactericidal activity even at concentrations as low as 0.625 µg/ml.

Structure-based homology analysis revealed that PHAb10 and PHAb11 probably belonged to the T4 lysozyme family and possessed the conserved Glu-Asp-Thr catalytic triad shared by traditional phage lysins and bacterial autolysins, namely Glu17, Asp26, and Thr32 of PHAb10, and Glu52, Asp61, and Thr67 of PHAb11. However, unlike most other enzymes reported to date, PHAb10 and PHAb11 exhibited excellent thermostability, retaining almost 100% bactericidal activity after treatment at 100°C for 1 hour. In contrast, gp36C lysin from bacteriophage φKMV, a known highly thermostable lysozyme, retains only ~50% residual activity after 1 hour of treatment at 100°C (46 [↗](#), 47 [↗](#)). Also, T4 lysozyme derived from T4 phage, which shares 44% sequence similarity and 27% sequence identity with gp36C, was completely inactivated after treatment at 75°C for 5 minutes (48 [↗](#)). To the best of our knowledge, phage-derived lysins with such high thermostability similar to PHAb10 and PHAb11 had yet to be described. Heat-resistant lysin have potential applications in animal feed premixes (pelletizing temperature 75-98°C) and medical devices requiring autoclaving.

To the best of our knowledge, few proteins have been demonstrated to have reversible folding-refolding thermodynamics at high temperature, but their mechanisms are unclear, such as HPL118, HPL511, and HPLP35 (49 [↗](#)). Here we found that PHAb10 had a unique temperature-dependent folding-refolding thermodynamic, and structural and biochemical analyses further suggested that PHAb10 undergoes a unique dimer-monomer switching mediated by 6 pairs of intermolecular interactions to survive heat treatment. To our knowledge, this is the first time that an enzyme had been found to adapt to ambient temperature by altering its dimerization state.

Bacteriophages are known to be the most abundant biological entities on earth (50 [↗](#)). Over billions of years of coevolution, more than 80% of bacterial genomes have become populated with at least one prophage (51 [↗](#), 52 [↗](#)), also known as molecular imprints, which serve as an untapped arsenal for the discovery of phage-related antimicrobial agents. Notably, prophage identification lags greatly behind genome or metagenome sequencing. Therefore, there is reason to believe that advances in bacterial culturomics (53 [↗](#)), ‘rebooting’ phage genomes in more tractable hosts (54 [↗](#)), and computational biology (55 [↗](#), 56 [↗](#)), may help discover otherwise inaccessible prophages from bacterial genome big data. In turn, this will increase the added value of bacterial genome as a new source of unearthed therapeutically promising antimicrobial drugs. More importantly, our findings implied that daily updated big data, such as bacterial genomes and proteomes, human proteomes, and various metagenomic data, could in principle become an encrypted power in the fight against antimicrobial resistance.

Materials and Methods

Bacterial strains

The bacterial strains used in this work are described in **Table S6**. *Acinetobacter baumannii*, *Escherichia coli*, *Pseudomonas aeruginosa*, *Klebsiella pneumoniae*, *Enterococcus faecalis*, and *Staphylococcus aureus* were all grown in Lysogeny Broth (LB) at 37°C. *Streptococcus pneumoniae* was stationary cultured in Todd Hewitt Broth with 0.5% Yeast extract (THY) at 37°C with 5% CO₂. All other *Streptococci* were grown in Brain Heart Infusion (BHI) broth at 37°C.

Screening of putative peptidoglycan hydrolases from *A. baumannii*

Putative peptidoglycan hydrolases from the *A. baumannii* (PHABs) proteome database were obtained by BLASTP in NCBI using the lysin PlyF307-derived antimicrobial peptide P307 as input sequence. Sequences marked as partial, shorter than 110 amino acids, longer than 300 amino acids, or clearly not belonging to lysins were manually excluded from further analysis. Multiple sequence alignments were performed by the ClustalW algorithm in MEGAX software and visualized by Jalview. Phylogenetic tree was further constructed by MEGAX software using the Neighbor-Joining method and visualized by the iTOL online service (<http://itol.embl.de/>). Three-dimensional (3D) protein models showing different amino acid residues were predicted online by SWISS-MODEL (<https://swissmodel.expasy.org/>).

In silico analysis of PHABs

The basic physical and chemical properties of PHABs were analyzed by ProtParam (<https://web.expasy.org/protparam/>). Solubility was predicted by Protein-Sol (<https://protein-sol.manchester.ac.uk/>), and signal peptides were predicted by SignalP (<https://services.healthtech.dtu.dk/service.php?SignalP-5.0>). The transmembrane area was predicted by TMHMM (<https://services.healthtech.dtu.dk/service.php?TMHMM-2.0>), while domain predictions were performed through CD searches in the NCBI database (<https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi>) and InterPro in the EBI database (<http://www.ebi.ac.uk/interpro/search/sequence/>). The Percent Identity Matrix was calculated by Clustal Omega (<https://www.ebi.ac.uk/Tools/msa/clustalo/>) and visualized with TBtools. The prophage origination analysis was predicted by PhageBoost (<https://phageboost.ku.dk/>) (57). The net charge of the N-terminal and C-terminal 30 amino acids of each PHAB was calculated by BaAMPs (<http://www.baamps.it/tools/calculator>) under neutral condition. The Local Net Charge of each PHAB was obtained by assigning 1 to the basic amino acids Arg (R) and Lys (K) and -1 to the acidic amino acids Glu (E) and Asp (D) and summing all assignments in a 31 amino acid window composing of 15 amino acids before and after the determined position. The 3D models of each PHAB were predicted by AlphaFold2 and visualized by PyMOL.

Gene synthesis and cloning

The coding sequences of the original five putative peptidoglycan hydrolases, namely, PHAB7, PHAB8, PHAB9, PHAB10, and PHAB11, as well as the PHAB10 variant, PHAB10-mut with 6 site mutations were chemically synthesized with codon optimization and cloned into *Nco*I and *Xho*I restriction sites of the pET28b(+) vector. Additional variants of PHAB10 and truncations of PHAB10 and PHAB11 were constructed by primer-derived overlap PCR (**Table S7**). All constructs were electro-transformed into *E. coli* BL21(DE3) competent cells for expression. Peptides derived from PHAB10 and PHAB11, namely P10-CP (amino acids 111-149), P11-NP (amino acids 1-36), and P11-CP (amino acids 146-184), were chemically synthesized and dissolved in ultrapure water before use.

Protein purification

BL21(DE3) cells were grown in LB to an optical density at 600 nm (OD_{600}) of 0.4–0.6 and induced with 0.2 mM isopropyl β -D-thiogalactopyranoside at 37°C for 2 h. Cells were then harvested and lysed by sonication. Supernatant was collected and passed through a Ni-NTA column pre-equilibrated with 20 mM imidazole. Fractions were collected by washing and eluting with 60 mM and 250 mM imidazole, respectively. The collected active protein fractions were pooled, dialyzed against 20 mM Tris-HCl (pH 7.4), and then filter-sterilized. Protein for crystallization was further purified by size-exclusion chromatography on a Superdex 75 (GE Healthcare) with gel filtration buffer (50 mM Tris-HCl, pH 8.0, 150 mM NaCl, 1 mM DTT). Protein purity was examined by 12% SDS-PAGE, and protein concentration determined using the BCA Protein Concentration Assay Kit. Dimerization of PHAb10 and its variants was detected by Native-PAGE.

Halo assay

Bacterial strains were grown overnight in 100 ml LB, harvested, and resuspended in 50 ml phosphate buffered saline (PBS) containing 0.7% agarose. Cells were then autoclaved and poured into glass Petri dish culture plates. Ten microliter of peptidoglycan hydrolases at different concentrations (0, 2.5, 5, and 10 μ g/ml) were dropped onto each bacterial lawn and incubated at 37°C till a clear zone was formed, and its area was determined using the method previously described (58). Wells treated with an equal volume of 20 mM Tris-HCl were used as controls.

Antibacterial activity assay

To determine the antibacterial activity of peptidoglycan hydrolases, bacteria were cultured overnight (stationary phase, OD_{600} = 1.2–1.6), then transferred to fresh medium at 100-fold dilution and continued to grow for 3 hours to OD_{600} = 0.5–0.6 in the exponential growth phase. Cultures of different bacteria were centrifuged at $10,000 \times g$ for 1 minute, washed once and resuspended in 20 mM Tris-HCl, pH 7.4. Stationary phase bacterial cells were diluted to a final OD_{600} of 0.6 before use. To test the bactericidal activity of each peptidoglycan hydrolase, 100 μ l of a bacterial suspension in stationary phase or exponential phase was mixed with an equal volume of peptidoglycan hydrolase at different concentrations at 37°C for 1 hour with shaking at 200 rpm. The number of viable bacteria after each treatment was counted by plating serial dilutions on LB agar. Wells treated with an equal volume of dialysis buffer instead of peptidoglycan hydrolases were used as controls. Specifically, to test the dose-dependent antibacterial activity of PHAb10 and PHAb11, exponential cultures of various Gram-negative and Gram-positive bacteria were treated with 0, 0.625, 1.25, 2.5, 5, 6.25, 10, 12.5, 25, 50, or 100 μ g/ml of each enzyme in 20 mM Tris-HCl (pH 7.4) at 37°C for 1 hour.

To test the time-killing curves of PHAb10 and PHAb11 against exponential *A. baumannii* 3437, bacterial suspensions were incubated with 50 μ g/ml of each peptidoglycan hydrolase in 20 mM Tris-HCl (pH 7.4) for 0, 0.5, 1, 3, 5, 10, 15, 30, 45, 60, and 120 min at 37°C.

To measure the effects of different environmental factors on the antibacterial activity of PHAb10 and PHAb11, exponential *A. baumannii* 3437 cells were treated with 50 μ g/ml of each peptidoglycan hydrolase at different pH values (5, 6, 7, 8, 9, and 10), various temperatures (4, 25, 37, 45, 55, 65, 75, 85, and 100°C), and various NaCl concentrations (0, 25, 50, 100, 200, 500, and 1000 mM), and urea (0, 50, 100, 200, 500, and 1000 mM) at 37°C for 1 hour. All assays were performed at least three times in biological replicates.

Thermostability assay

The thermostability of PHAb and their variants at 100 μ g/ml was determined by a Prometheus NT.48 nano-differential scanning fluorimetry (nanoDSF, NanoTemper Technologies, CA, USA) in the temperature range of 20–100°C (increasing step of 1°C/min) in 20 mM Tris-HCl (pH 7.4). The

first derivative of the fluorescence ratio at 350 and 330 nm (1st der, F350/F330) was calculated automatically by the PR-ThermControl software supplied with the instrumentation. The transition temperature (T_m) corresponds to the 1st order peak of F350/F330.

Circular dichroism

The circular dichroism spectra of PHAb and its variants at 200 $\mu\text{g/ml}$ in 20 mM Tris-HCl (pH 7.4) were collected by an Applied Photophysics Chirascan Plus circular dichroism spectrometer (Leatherhead, UK) from 200–260 nm (0.1 cm path length) at room temperature. The spectra of air and Tris-HCl buffer (pH 7.4) were recorded as background and baseline, respectively. Secondary structures were calculated by CDNN V2.1 software supplied by the instrument manufacturer.

Crystallization, data collection and processing

In this study, crystals were obtained for each protein at a concentration of 20–25 mg/ml using sitting drop method at a 1:1 ratio on a reservoir of 0.1 M citric acid pH 3.5, 25% w/v polyethylene glycol 3350 after 1 week of incubation at 16°C. They were then immersed in a cryoprotectant consisting of a reservoir solution supplemented with 10% glycerol before being flash frozen in liquid nitrogen. Diffraction data were collected at beamline 19U1 (BL19U1) at Shanghai Synchrotron Radiation Facility (SSRF), and the X-ray datasets were processed using HKL2000 and XDS software programs. The initial phase information was determined by molecular replacement in the program PHASER using *A. baumannii* AB 5075UW prophage (PDB code: 6ET6) as the initial search model. The structure was then improved by multiple rounds of manual construction and refinement by COOT and PHENIX. The final structures of PHAb8, PHAb10, and PHAb11 were visualized using PyMOL and deposited in the PDB database with access numbers 8HEM, 7YKU, and 8HDQ, respectively. Summary of data collection and refinement statistics are presented in [Table S2](#) [↗](#).

Cytotoxicity testing

The cytotoxicity of PHAb10 against HepG2 cells was determined by a Cell Counting Kit-8 (CCK-8) assay (Dojindo Molecular Technologies, Kumamoto, Japan) according to the manufacturer's protocol. HepG2 cells were seeded in 96-well plates at a density of 5,000 cells per well in Dulbecco's modified Eagle medium (DMEM; Sigma-Aldrich, Shanghai, China) supplemented with 10% fetal bovine serum, 1% penicillin, and 1% streptomycin for 24 h. The cells were then exposed to a series of concentrations of PHAb10 (0, 6.25, 12.5, 25, 50, and 100 $\mu\text{g/ml}$) for another 24 h. Afterwards, the contents of the plates were replaced with fresh medium containing 10% CCK-8 solution and incubated at 37°C for 1.5 h. The final optical density at OD450 was noted by use of a microplate reader (SynergyH1; BioTek, USA). The results were expressed as relative cell viability, expressed as a percentage of the growth of cells in control wells treated with PBS only.

Mouse experiments

All mouse infection experiments were conducted in an ABSL-2 lab, and all experimental methods were carried out in accordance with the regulations and guidelines set forth by the Animal Experiments Committee of the Wuhan Institute of Virology, Chinese Academy of Sciences. All experimental protocols were approved by the Animal Experiments Committee of Wuhan Institute of Virology, Chinese Academy of Sciences (WIVA17202102). Animals were housed in individually ventilated cages following a set of animal welfare and ethical criteria during the experiment and were euthanized at the end of the observation. In the burn infection model, 6- to 8-week-old female BALB/c mice were anesthetized, shaved, and depilated to create a 2-cm² hairless area on the dorsum as previously described ([35](#) [↗](#)). Partial thickness burn was achieved by exposing bare skin to water at 65°C for 12 s. The burns were then inoculated with 10 μl 5×10^8 cfu/ml of *A. baumannii* 3437. 24 hours after colonization, mice were randomly separated into four groups and treated with 5 or 10 $\mu\text{g/mouse}$ PHAb10 (10 μl ; $n = 5$), 4 $\mu\text{g/mouse}$ minocycline (10 μl ; $n = 5$), or an equal volume of Tris buffer (10 μl ; $n = 5$). Four hours after treatment, mice were sacrificed by

cervical dislocation to obtain skin. Specifically, the infected skin was excised, infiltrated in 1ml PBST (PBS containing 0.1% Triton X-100) for 2 min, and homogenized using a NewZongKe MD1000 Tissue Cell-Destroyer (NewZongKe, Wuhan, Hubei, China). The resulting solution was serially diluted and plated on LB agar containing 4 µg/ml gentamicin and 2 µg/ml meropenem. Viable cell numbers in each group were analyzed by one-way analysis of variance (ANOVA).

In the mouse abscess model, 6- to 8-week-old female mice were depilated to create a 2-cm² hairless area on the dorsum as described above and infected hypodermically (i.h.) with 25 µl of *A. baumannii* 3437 at a concentration of 5×10^8 cfu/ml on the right side of the back dorsum. After 24 h of infection, mice were randomly divided into 3 groups and hypodermically treated with 10 µg/mouse PHAb10 (10 µl; n = 6), 4 µg/mouse minocycline (10 µl; n = 7), or an equal volume of Tris buffer (10 µl; n = 8). Subcutaneous dosage was performed once a day for 5 consecutive days for each group. The body weight of mice in each group was monitored daily. Twenty-four hours after the last dosage, mice were sacrificed by cervical dislocation. Infected skin was excised, infiltrated in 1ml PBST for 2 min, and subjected to tissue homogenization using a NewZongKe MD1000 Tissue Cell-Destroyer. The resulting solution was serially diluted and plated on LB agar containing 4 µg/ml gentamicin and 2 µg/ml meropenem. The number of viable bacteria in each group was analyzed by ANOVA.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (No. 32070187, 32161133003, 31770192, and 81802001), and the Open Research Fund Program of National Bio-Safety Laboratory, Wuhan (No. 2021SPCAS001). We thank the staff of BL19U1 beamline at National Center for Protein Sciences Shanghai and Shanghai Synchrotron Radiation Facility (Shanghai, China) for assistance during data collection. We thank Dr. Xuefang An, and Dr. Yanfeng Yao from the Core Facility and Technical Support, Wuhan Institute of Virology for their assistance in animal experiments.

Author contributions

Li Zhang: Investigation; Methodology; Writing—review and editing. **Fen Hu:** Investigation; Methodology; Funding acquisition; Writing—original draft; Writing—review and editing. **Zirong Zhao:** Investigation; Methodology; Writing—review and editing. **Xinfeng Li:** Investigation; Methodology; Writing—review and editing. **Mingyue Zhong:** Investigation; Methodology; Writing—review and editing. **Jiajun He:** Investigation; Methodology; Writing—review and editing. **Fangfang Yao:** Investigation; Methodology; Writing—review and editing. **Xiaomei Zhang:** Methodology; Writing—review and editing. **Yuxuan Mao:** Methodology; Writing—review and editing. **Hongping Wei:** Supervision; Methodology; Writing—review and editing. **Jin He:** Conceptualization; Supervision; Methodology; Writing—review and editing. **Hang Yang:** Conceptualization; Supervision; Investigation; Methodology; Methodology; Funding acquisition; Writing—original draft; Writing—review and editing.

Competing interests

The authors declare no competing financial interests.

Supplemental Materials

Parameter	Amino acid sequence	Molecular weight (kDa)	pI	Negatively /positively charged residues	Net charge	Instability index	Aliphatic index	GRAVY
P10-CP	NKGDYQ GAADQFL VWNKAG GKVMKGL VRRREAE RALFLKK	4550	10.3 7	4/10	6	39.66	73.25	-0.733
P11-CP	NKGDYKG AADQFLV WNKAGG KVLKGLV RRREAER ALFLKK	4532	10.4 5	4/11	7	35.41	83.00	-0.695
P11-NP	MTTKPFF DAARVIA GGKLTQA QVDELNK VVNKLAP G	3829	9.52	3/5	2	19.74	89	-0.053

Positively charged amino acids are labeled in red.

GRAVY: Grand average of hydropathicity.

Table S1.

Physicochemical properties of putative antimicrobial peptides in PHAb10 and PHAb11

Parameters	PHAb10	PHAb11	PHAb8
Data collection			
Wavelength (Å)	0.9793	0.9793	0.9793
Space group	<i>P</i> 1 2 ₁ 1	<i>P</i> 2 2 2 ₁	<i>C</i> 2 2 2 ₁
Cell dimensions			
a, b, c (Å)	31.86 73.77 78.62	50.01 54.9 161.31	79.03 156.98 47.26
α, β, γ (°)	90.00 93.02 90.00	90 90 90	90 90 90
Resolution range (Å)	31.82 - 1.52 (1.574 - 1.52)	36.62 - 2.27 (2.351 - 2.27)	32.06 - 2.4 (2.486 - 2.4)
Unique reflections	55245 (5468)	21246 (2049)	11880 (1048)
Multiplicity	6.8 (6.5)	12.3 (12.4)	12.9 (13.2)
Completeness (%)	98.86 (98.17)	99.65 (99.80)	94.57 (88.22)
Mean I/sigma I	18.36 (1.67)	14.54 (2.02)	10.50 (1.53)
Wilson B-factor (Å ²)	24.27	31.87	37.86
R _{merge}	0.04816 (1.097)	0.2869 (1.486)	0.213 (1.914)
R _{meas}	0.0522 (1.192)	0.2999 (1.55)	0.2219 (1.991)
CC _{1/2}	0.999 (0.788)	0.992 (0.747)	0.996 (0.579)
Refinement			
Resolution (Å)	31.82 - 1.52	36.62 - 2.27	32.06 - 2.4
R _{work}	0.1921 (0.3344)	0.2483 (0.3092)	0.2520 (0.3560)
R _{free}	0.2151 (0.3653)	0.2660 (0.3670)	0.2849 (0.4396)
Ligands	0	18	17
Rmsd from ideal			
Bond lengths (Å)	0.004	0.005	0.010
Bond angles (°)	0.62	0.75	1.18
Ramachandran plot			
Favored (%)	98.97	98.25	95.81
Allowed (%)	1.03	1.75	4.19
Outliers (%)	0.00	0.00	0.00
Average B-factor	32.18	37.23	52.84

Table S2.

Data collection and refinement statistics of PHAb8, PHAb10, and PHAb11 structures

Table S2. (continued)

PDB Code

7YKU

8HDQ

8HEM

Statistics for the highest-resolution shell are shown in parentheses.

Table S3.

T4L-like lysozymes with characterized structures

Protein	Catalytic Residues			PDB ID
PHAb8	Glu74	Asp83	Thr89	8HEM
PHAb10	Glu17	Asp26	Thr32	7YKU
PHAb11	Glu52	Asp61	Thr67	8HDQ
AcLys	Glu64	Asp73	Thr79	6ET6
LysF1	Glu15	Asp24	Thr30	7M5I
SpmX-Mur- <i>Ae</i>	Glu18	Leu27	Met33	6H9D
P22	Glu16	Asp25	Thr31	2ANX
R21	Glu35	Asp44	Thr50	3HDE
HEWL	Glu35	Asp52		4HPI
GEWL	Glu73			153L
phage lambda lytic transglycosylase	Glu19			1D9U
T4L	Glu11	Asp20	Thr26	1LYD

Table S4.

Intermolecular interactions of PHAb10 dimer

PHAB10 dimer	Serial number	Donor (atom 1)	Acceptor (atom 2)	Distance/Å	Force
Head-supporting force	1	R20.NH1 (Chain A)	G129.O (Chain B)	3.5	H-bond
	2	T104.N (Chain B)	G27.O (Chain A)	3.1	H-bond
	3	T104.OG1 (Chain B)	G27.O (Chain A)	3.5	H-bond
Midbody-supporting force	4	E97.N (Chain B)	G128.O (Chain A)	2.9	H-bond
	5	K67.NZ (Chain B)	G129.O (Chain A)	2.9	H-bond
Tail-supporting force	6	K133.NZ (Chain A)	D26.O (Chain B)	2.9	H-bond
	7	R137.NH1 (Chain A)	G27.O (Chain B)	2.7	H-bond

Table S5.

PHAb10 variants

Native amino acids (Chain A)	Substituted amino acids	Mutant name
R20, G27	R20I, G27D	R20I/G27D
G128, G129	G128D, G129D	G128D/G129D
G27, R137	G27D, R137I	G27D/R137I
K133, R137	K133A, R137I	K133A/R137I
R20, G27, K67, G128, K133, R137	R20E, G27T, K67W, G128N, K133A, R137N	PHAb10-mut

Strain	Source	Strain	Source
<i>Acinetobacter baumannii</i>		<i>Klebsiella pneumoniae</i>	
ABA3437	WIV ^b	KP21	WIV ^b
CRAB2	WIV ^b	KP22	WIV ^b
CRAB13	WIV ^b	KP23	WIV ^b
CRAB36	WIV ^b	KP24	WIV ^b
CRAB27	WIV ^b	KP26	WIV ^b
CRAB28	WIV ^b	KP27	WIV ^b
CRAB22	WIV ^b	KP28	WIV ^b
CRAB26	WIV ^b	KP602	WIV ^b
CRAB140	WIV ^b	KP605	WIV ^b
CRAB143	WIV ^b	KP607	WIV ^b
CRAB147	WIV ^b	<i>Pseudomonas aeruginosa</i>	
<i>Escherichia coli</i>		PAE2555	WIV ^b
<i>E. coli</i> 100	WIV ^b	PAE1	WIV ^b
<i>E. coli</i> BL21	WIV ^b	PAE2	WIV ^b
<i>E. coli</i> O91	WIV ^b	PAE52	WIV ^b
<i>E. coli</i> O97	WIV ^b	PAE53	WIV ^b
<i>E. coli</i> O149	WIV ^b	PAE2139	WIV ^b
<i>E. coli</i> O517	WIV ^b	PAE2169	WIV ^b
<i>E. coli</i> MS31	WIV ^b	PAE2176	WIV ^b
<i>E. coli</i> MS33	WIV ^b	PAE2703	WIV ^b
<i>E. coli</i> MS28	WIV ^b	PAE3223	WIV ^b
<i>E. coli</i> MS34	WIV ^b	PA2-3121	WIV ^b
<i>Staphylococcus aureus</i>		PA2-3283	WIV ^b
<i>S. aureus</i> N315	WIV ^b	PA2-3763	WIV ^b
<i>S. aureus</i> RN4220	WIV ^b	PA2-3965	WIV ^b
<i>S. aureus</i> NF71	WIV ^b	<i>Streptococcus</i>	
<i>S. aureus</i> 25923	WIV ^b	<i>S. pneumoniae</i> 17NS26	HPPH ^a
<i>Enterococcus faecalis</i>		<i>S. agalactiae</i> 12	WIV ^b
<i>E. faecalis</i> 493	WIV ^b	<i>S. agalactiae</i> C001	WIV ^b
<i>E. faecalis</i> 29212	ATCC ^c	<i>S. dysgalactiae</i> 35666	ATCC ^c
<i>E. faecalis</i> 51299	WIV ^b	<i>S. pyogenes</i> 12344	ATCC ^c
<i>E. faecalis</i> 571-2	WIV ^b	<i>S. mutans</i> 159	WIV ^b
		<i>S. Suis</i> 1	HUAHE ^d

a: Isolated from Henan Provincial People's Hospital, Zhengzhou, China.

b: Obtained from Wuhan Institute of Virology, Chinese Academy of Sciences, Wuhan, China.

c: Purchased from Guangdong Culture Collection Center, China.

d: Obtained from Henan University of Animal Husbandry and Economy, Zhengzhou, China.

Table S6

Bacterial strains used in this work.

Primer	Sequence (5'-3')
P10-Lys-F	ctttaagaaggagatataaccATGgcatgagcaaaaccaccagt
P10-Lys-R	gtggtggtgctcgagCAGtttttcagcaggg
P11-N-F	agaaggagatataaccATGgcatgaccaccaaacggtttttgatgcag
P11-N-R	gtggtggtgctcgagCAGcagtttcagcagggtgcta
P11-C-F1	aactttaagaaggagatataccatgggtggtaaaaccacc
P11-C-R1	gggtggtggtgctcgagtttttcagaaacagtgcacg
P11-C-F2	aactcgagcaccaccaccaccac
P11-C-R2	ggtatatctccttcttaaagtaaacaaaaattttctagag
K133AR137I-F1	tttgttaactttaagaaggagatataccatgggcatgagcaaaaccacc
K133AR137I-R1	acgacggataaccagaccggccataactttaccacc
K133AR137I-F2	tggccgggtctggttatccgtcgtgaagcagaac
K133AR137I-R2	gctcatgcccatggtatatctccttcttaaagtaaacaaaaattttc
R20IG27D-F1	accaccagtaatgcaggctgaatctgatcaaagggtttgaaggtaaaatcctgaatgc
R20IG27D-R1	gtcccaaaaccaatggccaacacacatcatcatcataggcattcaggattttacc
R20IG27D-F2	ccattgggtttggcaccatcaaatccgaatgggttcg
R20IG27D-R2	cctgcattactgggtgtttgctcatgcccatggtatatc
G128DG129D-F1	atgagcaaaaccaccagtaatgcaggctgaatctgatcaaag
G128DG129D-R1	aactttatcatcggtttattccaaaccagaaactgatctgctgc
G128DG129D-F2	aagccgatgataaagttatgaaaggctcgggtcgtcgtcg
G128DG129D-R2	actggtggttttgcctcatgcccatggtatatctccttc
G27DR137I-F1	tgatgatgttggtgtttggaccattggttttggcacc
G27DR137I-R1	tctgcttcacgacggataaccagaccttcataactttaccac
G27DR137I-F2	ggttatccgtcgtgaagcagaacgtgcactg
G27DR137I-R2	ccaaacaccaacatcatcatcataggcattcaggcg

Table S7.

Primers used in this study.

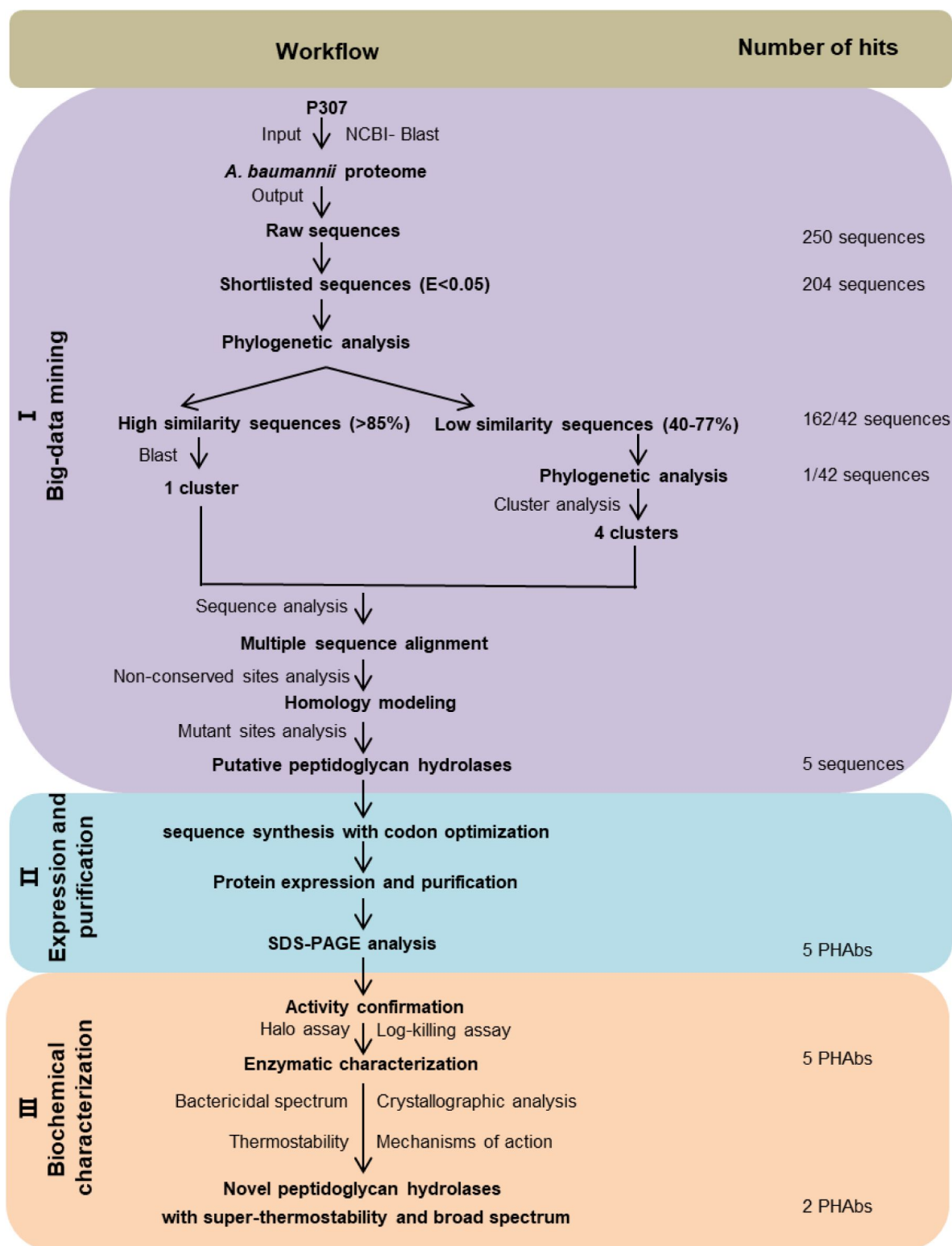


Figure S1.

Workflow of an integrated system for the exploitation of peptidoglycan hydrolases.

Three distinct phases of the workflow were highlighted in different colors: (I) Screening of peptidoglycan hydrolases using cationic peptides P307 as templates (yellow). (II) Expression and purification of protein (blue). (III) Biochemical characterization (green).

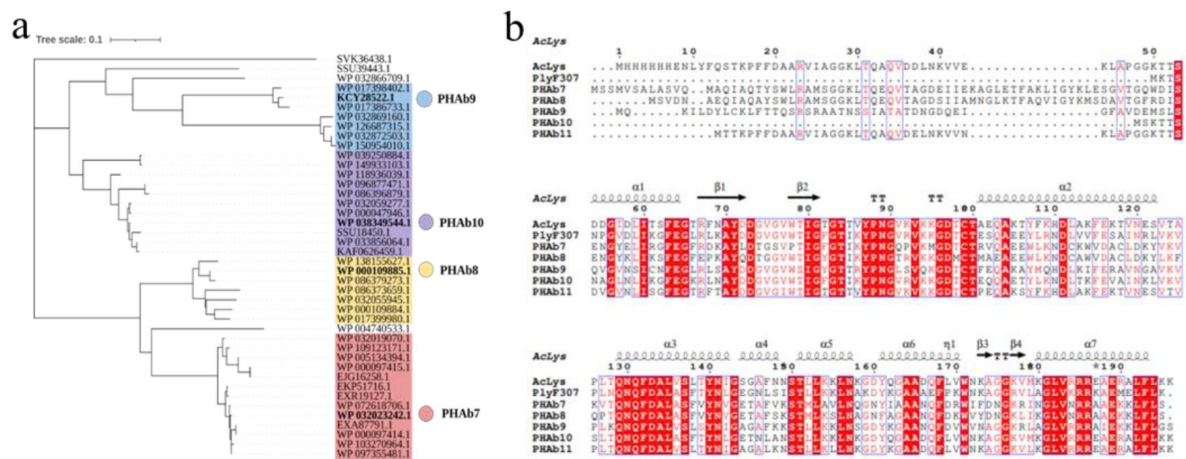


Figure S2.

Selection of five putative peptidoglycan hydrolases.

(a) Phylogenetic tree of *A. baumannii* peptidoglycan hydrolases homolog to P307. (b) Multiple sequence alignment of PHAb7, PHAb8, PHAb9, PHAb10, PHAb11, PlyF307, and AcLys using MEGAX software and ClustalW algorithm. Secondary structures were annotated by Esript 3.0 website.

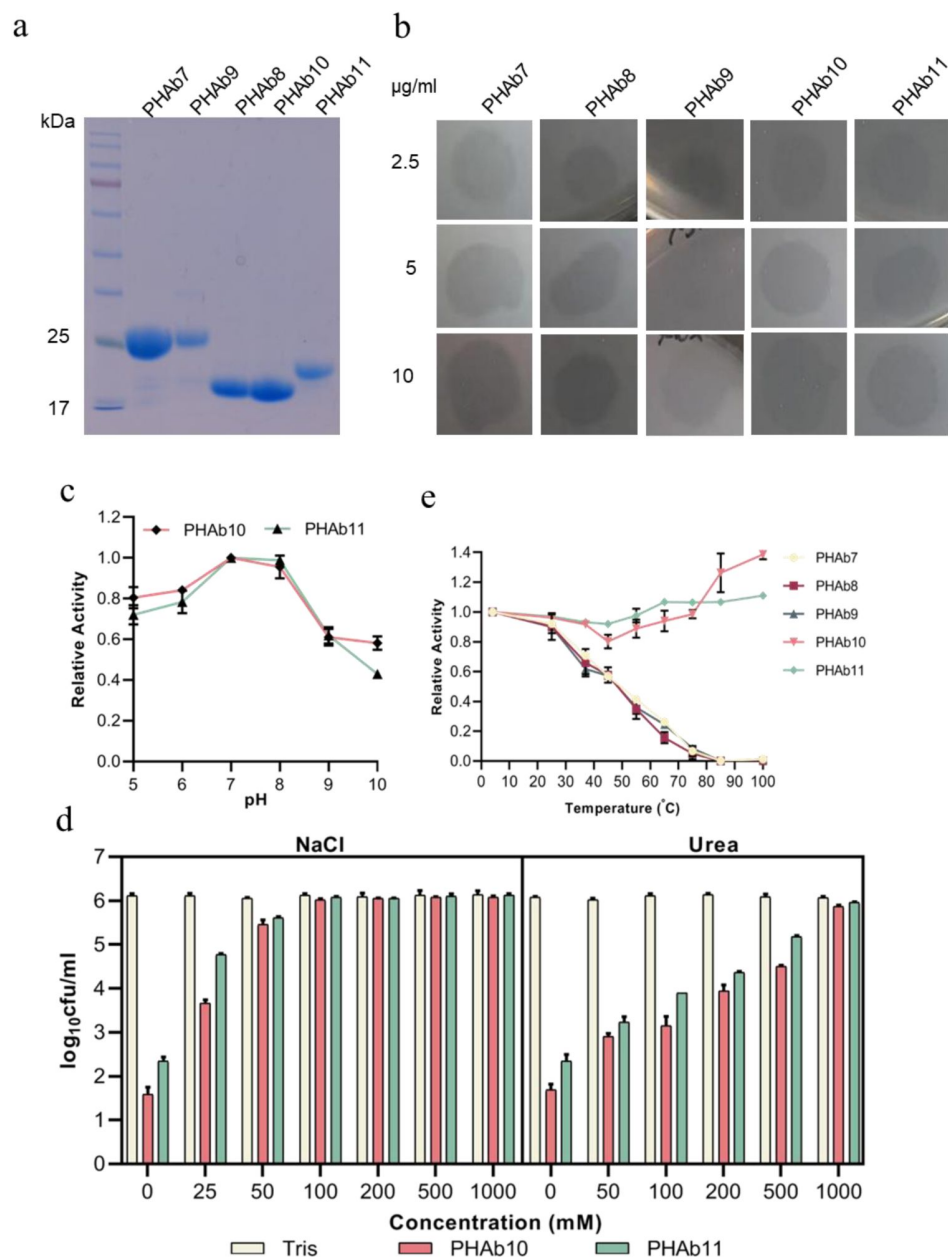


Figure S3.

Testing of bactericidal activity of screened putative peptidoglycan hydrolases.

(a) Purified proteins were analyzed on a 12% SDS-PAGE gel. M: standard protein marker. (b) Halos of five PHAbs on *A. baumannii* 3437 lawns. The concentrations of each peptidoglycan hydrolase were 2.5 µg/ml, 5 µg/ml, and 10 µg/ml. (c) pH stability of PHAb10 and PHAb11. Log-phase *A. baumannii* 3437 cells were incubated with 50 µg/ml peptidoglycan hydrolases at the indicated pH conditions for 1 hour at 37°C. pH buffer: 20 mM Boric acid mixed with 20 mM phosphoric acid in equal volumes and adjusted to different pH values with HCl. (d) Log-phase *A. baumannii* 3437 cells incubated with 50 µg/ml of each peptidoglycan hydrolase at different concentrations of NaCl (left) and urea (right) for 1 hour at 37°C in 20 mM Tris-HCl (pH 7.4). (e) Relative activity of five PHAbs against *A. baumannii* 3437 after 1 hour storage at different temperatures. In all cases, viable bacteria were counted by serial dilution and plating assays. Experiments were carried out in triplicate; error bars represent standard deviation.

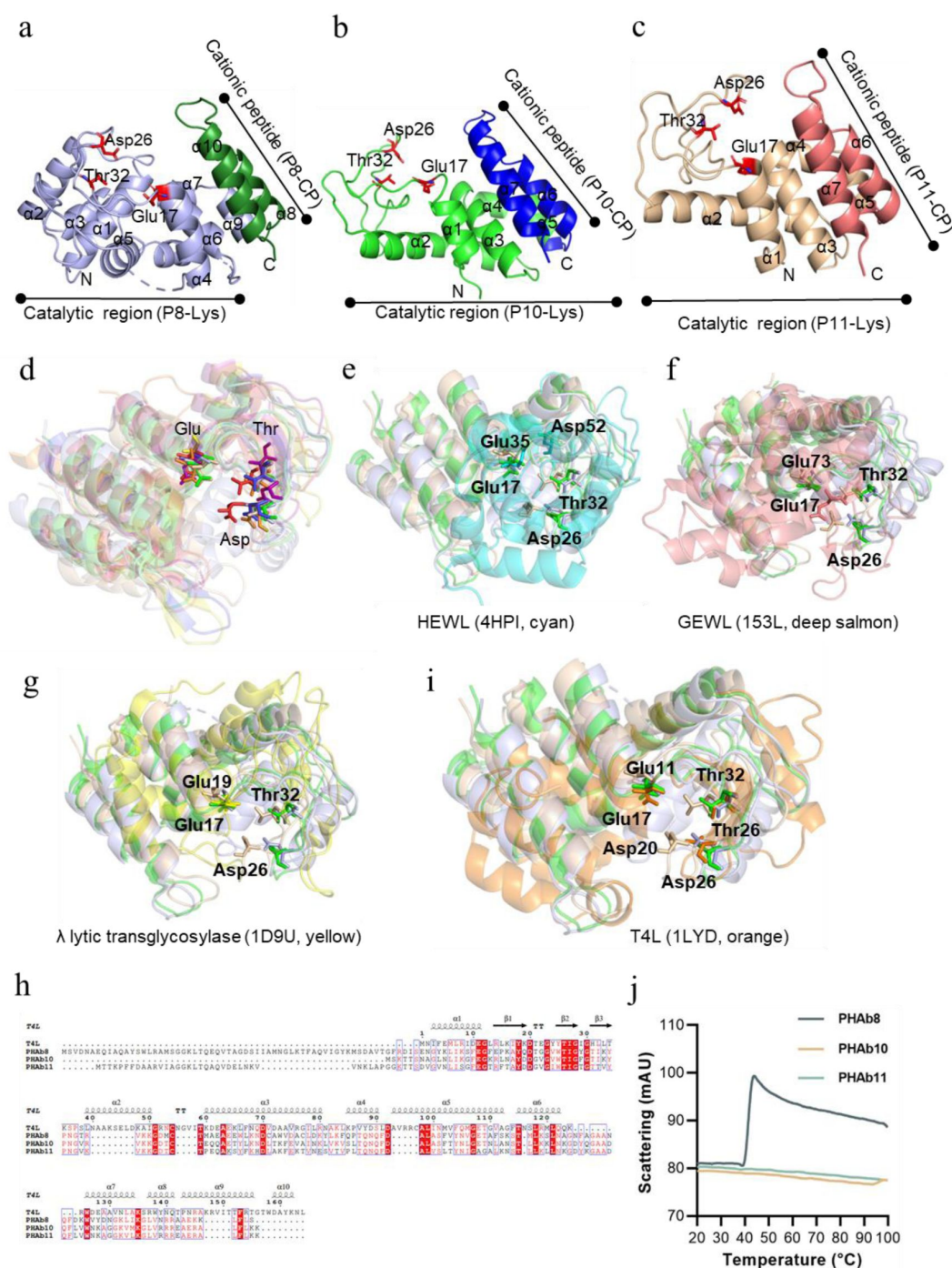


Figure S4.

Structural analysis of PHAb8, PHAb10, and PHAb11.

(a-c) Structures of monomeric PHAb8 (a), PHAb10 (b), and PHAb11 (c). (d) Structural superposition of PHAb8, PHAb10, PHAb11 and five classical T4L-like lysozymes that share high similarity from Dali search. PHAb8 is shown in light blue, PHAb10 in green, PHAb11 in wheat, Aclys in red, LysF1 in blue, SpmX-Mur-Ae in yellow, P22 in purple, and R21 in orange. (e-h) Structural alignment of PHAb8, PHAb10, PHAb11 with various lysozymes including HEWL (e, cyan), GEWL (f, deep salmon), phage lambda lytic transglycosylase (g, yellow) and T4 lysozyme (h, orange). Residues in the conservative catalytic triad are shown as sticks. (j) Scattering profiles of PHAb8, PHAb10, and PHAb11 at different temperatures determined by NanoDSF.

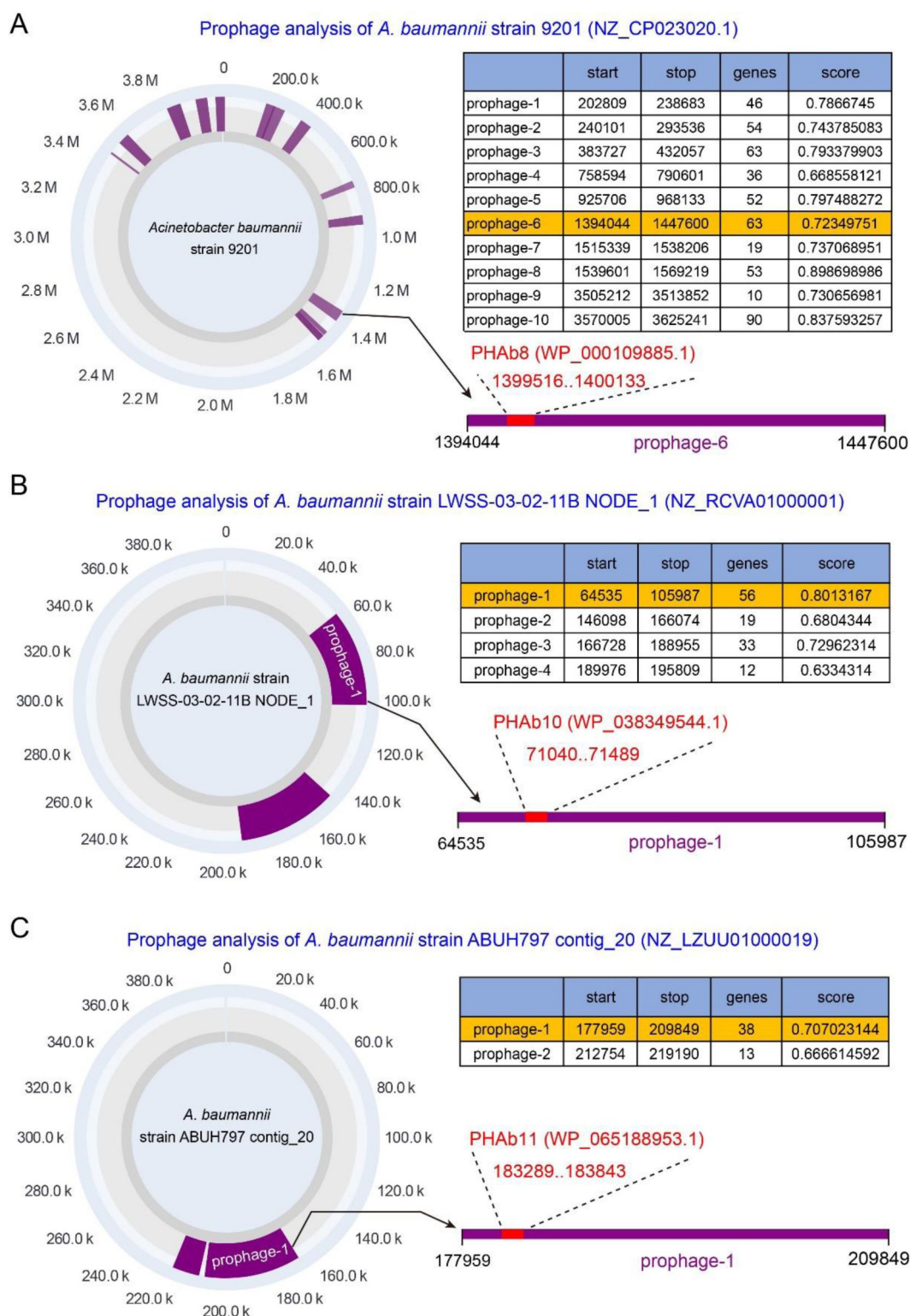


Figure S5.

Prophage analysis of PHAbs.

PHAb8, PHAb10, and PHAb11 encoding sequences were retrieved from *A. baumannii* strain 9201, *A. baumannii* strain LWSS-03-02-11B, and *A. baumannii* strain ABUH797 and analyzed by PhageBoost tool. Results revealed that all three genes are indeed located within prophage regions.

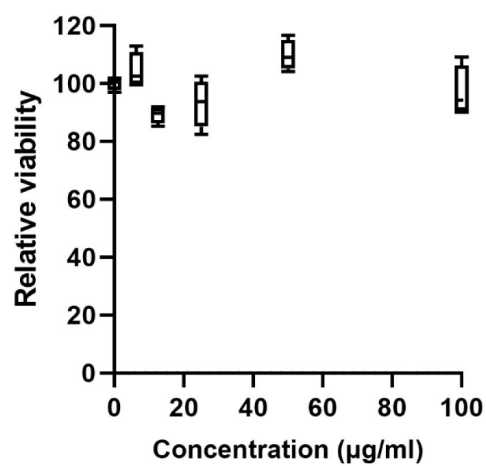


Figure S6.

Relative viability of HepG2 cells exposed to different concentrations of PHAb10.

Cells were cultured in DMEM supplemented with 10% fetal bovine serum, 1% penicillin, and 1% streptomycin for 24 h. After the cells were exposed to a series of concentrations of PHAb10 (0, 6.25, 12.5, 25, 50, and 100 µg/ml) for another 24 h, the relative viability of the cells after each treatment was determined by the CCK-8 assay.

References

1. Nathan C (2020) **Resisting antimicrobial resistance** *Nat Rev Microbiol* **18**:259–260
2. Mendelson M, Morris AM, Thursky K, Pulcini C (2020) **How to start an antimicrobial stewardship programme in a hospital** *Clin Microbiol Infect* **26**:447–453
3. Rawson TM, Ming D, Ahmad R, Moore LSP, Holmes AH (2020) **Antimicrobial use, drug-resistant infections and COVID-19** *Nat Rev Microbiol* **18**:409–410
4. Van Boeckel TP, Pires J, Silvester R, Zhao C, Song J, Criscuolo NG, Gilbert M, Bonhoeffer S, Laxminarayan R (2019) **Global trends in antimicrobial resistance in animals in low- and middle-income countries** *Science* **365**
5. McEwen SA, Collignon PJ (2018) **Antimicrobial Resistance: a One Health Perspective** *Microbiol Spectr* **6**
6. Hernando-Amado S, Coque TM, Baquero F, Martinez JL (2019) **Defining and combating antibiotic resistance from One Health and Global Health perspectives** *Nat Microbiol* **4**:1432–1442
7. Talaat M, Zayed B, Tolba S, Abdou E, Gomaa M, Itani D, Hutin Y, Hajjeh R (2022) **Increasing Antimicrobial Resistance in World Health Organization Eastern Mediterranean Region, 2017–2019** *Emerg Infect Dis* **28**:717–724
8. Brink AJ (2019) **Epidemiology of carbapenem-resistant Gram-negative infections globally** *Curr Opin Infect Dis* **32**:609–616
9. Theuretzbacher U, Outterson K, Engel A, Karlen A (2020) **The global preclinical antibacterial pipeline** *Nat Rev Microbiol* **18**:275–285
10. Lazzaro BP, Zasloff M, Rolff J (2020) **Antimicrobial peptides: Application informed by evolution** *Science* **368**
11. Silver LL (2016) **Appropriate Targets for Antibacterial Drugs** *Cold Spring Harb Perspect Med* **6**
12. Mayer RL, Impens F (2021) **Immunopeptidomics for next-generation bacterial vaccine development** *Trends Microbiol* **29**:1034–1045
13. Luong T, Salabarria AC, Edwards RA, Roach DR (2020) **Standardized bacteriophage purification for personalized phage therapy** *Nat Protoc* **15**:2867–2890
14. De Maesschalck V, Gutierrez D, Paeshuyse J, Lavigne R, Briers Y. (2020) **Advanced engineering of third-generation lysins and formulation strategies for clinical applications** *Crit Rev Microbiol* **46**:548–564
15. Korpela K, Helve O, Kolho KL, Saisto T, Skogberg K, Dikareva E, Stefanovic V, Salonen A, Andersson S, de Vos WM. (2020) **Maternal Fecal Microbiota Transplantation in Cesarean-Born Infants Rapidly Restores Normal Gut Microbial Development: A Proof-of-Concept Study** *Cell* **183**:324–334

16. Dickey SW, Cheung GYC, Otto M (2017) **Different drugs for bad bugs: antivirulence strategies in the age of antibiotic resistance** *Nat Rev Drug Discov* **16**:457–471
17. Bush K, Bradford PA (2019) **Interplay between beta-lactamases and new beta-lactamase inhibitors** *Nat Rev Microbiol* **17**:295–306
18. Yang H, Xu J, Li W, Wang S, Li J, Yu J, Li Y, Wei H (2018) **Staphylococcus aureus virulence attenuation and immune clearance mediated by a phage lysin-derived protein** *EMBO J* **37**
19. Federici S, Nobs SP, Elinav E (2021) **Phages and their potential to modulate the microbiome and immunity** *Cell Mol Immunol* **18**:889–904
20. Batalha IL *et al.* (2019) **Polymeric nanobiotics as a novel treatment for mycobacterial infections** *J Control Release* **314**:116–124
21. Farha MA, Brown ED (2019) **Drug repurposing for antimicrobial discovery** *Nat Microbiol* **4**:565–577
22. Jampilek J (2022) **Drug repurposing to overcome microbial resistance** *Drug Discov Today* **27**:2028–2041
23. Torres MDT, Melo MCR, Crescenzi O, Notomista E, de la Fuente-Nunez C. (2022) **Mining for encrypted peptide antibiotics in the human proteome** *Nat Biomed Eng* **6**:67–75
24. Ma Y *et al.* (2022) **Identification of antimicrobial peptides from the human gut microbiome using deep learning** *Nat Biotechnol* **40**:921–931
25. Wrighton KH (2018) **Antibacterial drugs: Discovering antibiotics through soil metagenomics** *Nat Rev Drug Discov* **17**:240–241
26. Povolotskaya IS, Kondrashov FA (2010) **Sequence space and the ongoing expansion of the protein universe** *Nature* **465**:922–6
27. Randall JR, DuPai CD, Davies BW (2022) **Discovery of Antimicrobial Peptide Macrocycles Through Bacterial Display** *Methods Mol Biol* **2371**:287–298
28. Das P *et al.* (2021) **Accelerated antimicrobial discovery via deep generative models and molecular dynamics simulations** *Nat Biomed Eng* **5**:613–623
29. Schmelcher M, Loessner MJ (2021) **Bacteriophage endolysins - extending their application to tissues and the bloodstream** *Curr Opin Biotechnol* **68**:51–59
30. Theuretzbacher U, Piddock LJV (2019) **Non-traditional Antibacterial Therapeutic Options and Challenges** *Cell Host Microbe* **26**:61–72
31. Briers Y, Lavigne R (2015) **Breaking barriers: expansion of the use of endolysins as novel antibacterials against Gram-negative bacteria** *Future Microbiol* **10**:377–90
32. Lai WCB, Chen X, Ho MKY, Xia J, Leung SSY (2020) **Bacteriophage-derived endolysins to target gram-negative bacteria** *Int J Pharm* **589**

33. Lood R, Winer BY, Pelzek AJ, Diez-Martinez R, Thandar M, Euler CW, Schuch R, Fischetti VA (2015) **Novel phage lysin capable of killing the multidrug-resistant gram-negative bacterium *Acinetobacter baumannii* in a mouse bacteremia model** *Antimicrob Agents Chemother* **59**:1983–91
34. Thandar M, Lood R, Winer BY, Deutsch DR, Euler CW, Fischetti VA (2016) **Novel Engineered Peptides of a Phage Lysin as Effective Antimicrobials against Multidrug-Resistant *Acinetobacter baumannii*** *Antimicrob Agents Chemother* **60**:2671–9
35. Li C *et al.* (2021) **Intrinsic Antimicrobial Peptide Facilitates a New Broad-Spectrum Lysin LysP53 to Kill *Acinetobacter baumannii* In Vitro and in a Mouse Burn Infection Model** *ACS Infect Dis* **7**:3336–3344
36. Kim S, Jin JS, Choi YJ, Kim J (2020) **LysSAP26, a New Recombinant Phage Endolysin with a Broad Spectrum Antibacterial Activity** *Viruses* **12**
37. Yu L, Frank O, Philippe M, Grégory R, Manuel EJ, Stefano M (2018) **In vitro characterization of PlyE146, a novel phage lysin that targets Gram-negative bacteria** *PLoS one* **13**
38. Ghose C, Euler CW (2020) **Gram-Negative Bacterial Lysins** *Antibiotics (Basel)* **9**
39. Guo M, Feng C, Ren J, Zhuang X, Zhang Y, Zhu Y, Dong K, He P, Guo X, Qin J (2017) **A Novel Antimicrobial Endolysin, LysPA26, against *Pseudomonas aeruginosa*** *Front Microbiol* **8**
40. Khan FM, Gondil VS, Li C, Jiang M, Li J, Yu J, Wei H, Yang H (2021) **A Novel *Acinetobacter baumannii* Bacteriophage Endolysin LysAB54 With High Antibacterial Activity Against Multiple Gram-Negative Microbes** *Front Cell Infect Microbiol* **11**
41. Briers Y, Walmagh M, Lavigne R (2011) **Use of bacteriophage endolysin EL188 and outer membrane permeabilizers against *Pseudomonas aeruginosa*** *J Appl Microbiol* **110**:778–85
42. Walmagh M, Boczkowska B, Grymonprez B, Briers Y, Drulis-Kawa Z, Lavigne R (2013) **Characterization of five novel endolysins from Gram-negative infecting bacteriophages** *Appl Microbiol Biotechnol* **97**:4369–75
43. Oliveira H, Thiagarajan V, Walmagh M, Sillankorva S, Lavigne R, Neves-Petersen MT, Kluskens LD, Azeredo J (2014) **A thermostable *Salmonella* phage endolysin, Lys68, with broad bactericidal properties against gram-negative pathogens in presence of weak acids** *PLoS One* **9**
44. Plotka M, Kapusta M, Dorawa S, Kaczorowska AK, Kaczorowski T (2019) **Ts2631 Endolysin from the Extremophilic *Thermus scotoductus* Bacteriophage vB_Tsc2631 as an Antimicrobial Agent against Gram-Negative Multidrug-Resistant Bacteria** *Viruses* **11**
45. Raz A, Serrano A, Hernandez A, Euler CW, Fischetti VA (2019) **Isolation of phage lysins that effectively kill *Pseudomonas aeruginosa* in mouse models of lung and skin infection** *Antimicrobial Agents and Chemotherapy*
46. Lavigne R, Briers Y, Hertveldt K, Robben J, Volckaert G (2004) **Identification and characterization of a highly thermostable bacteriophage lysozyme** *Cell Mol Life Sci* **61**:2753–9

47. Briers Y, Lavigne R, Plessers P, Hertveldt K, Hanssens I, Engelborghs Y, Volckaert G (2006) **Stability analysis of the bacteriophage KMV lysin gp36C and its putative role during infection** *Cellular and Molecular Life Sciences CMLS* **63**:1899–1905
48. Tsugita A, Inouye M, Terzaghi E, Streisinger G (1968) **Purification of Bacteriophage T4 Lysozyme** *Journal of Biological Chemistry* **243**
49. Schmelcher M, Waldherr F, Loessner MJ (2012) **Listeria bacteriophage peptidoglycan hydrolases feature high thermoresistance and reveal increased activity after divalent metal cation substitution** *Applied Microbiology and Biotechnology* **93**:633–643
50. Knowles B *et al.* (2016) **Lytic to temperate switching of viral communities** *Nature* **531**:466–70
51. Miller-Ensminger T, Garretto A, Brenner J, Thomas-White K, Zambom A, Wolfe AJ, Putonti C (2018) **Bacteriophages of the Urinary Microbiome** *J Bacteriol* **200**
52. Keen EC, Dantas G (2018) **Close Encounters of Three Kinds: Bacteriophages, Commensal Bacteria, and Host Immunity** *Trends Microbiol* **26**:943–954
53. Lagier JC, Dubourg G, Million M, Cadoret F, Bilen M, Fenollar F, Levasseur A, Rolain JM, Fournier PE, Raoult D (2018) **Culturing the human microbiota and culturomics** *Nat Rev Microbiol* **16**:540–550
54. Kilcher S, Studer P, Muessner C, Klumpp J, Loessner MJ (2018) **Cross-genus rebooting of custom-made, synthetic bacteriophage genomes in L-form bacteria** *Proc Natl Acad Sci U S A* **115**:567–572
55. Arndt D, Marcu A, Liang Y, Wishart DS (2019) **PHAST, PHASTER and PHASTEST: Tools for finding prophage in bacterial genomes** *Brief Bioinform* **20**:1560–1567
56. Song W *et al.* (2019) **Prophage Hunter: an integrative hunting tool for active prophages** *Nucleic Acids Res* **47**:W74–W80
57. Siren K, Millard A, Petersen B, Gilbert MTP, Clokie MRJ, Sicheritz-Ponten T. (2021) **Rapid discovery of novel prophages using biological feature engineering and machine learning** *NAR Genom Bioinform* **3**
58. Zhang BZ *et al.* (2017) **Immunotherapy Targeting Adenosine Synthase A Decreases Severity of Staphylococcus aureus Infection in Mouse Model** *J Infect Dis* **216**:245–253

Editors

Reviewing Editor

Bavesh Kana

University of the Witwatersrand, Johannesburg, South Africa

Senior Editor

Bavesh Kana

University of the Witwatersrand, Johannesburg, South Africa

Reviewer #1 (Public Review):**Summary:**

Li Zhang et al. characterized two new Gram-negative endolysins identified through an AMP-targeted search in bacterial proteomes. These endolysins exhibit broad lytic activity against both Gram-negative and Gram-positive bacteria and retain significant antimicrobial activity even after prolonged exposure to high temperatures (100°C for 1 hour). This stability is attributed to a temperature-reversible transition from a dimer to a monomer. The authors suggest several potential applications, such as complementing heat sterilization processes or being used in animal feed premixes that undergo high-temperature pelleting, which I agree with.

Strengths:

The claims are well-supported by relevant and complementary assays, as well as extensive bioinformatic analyses.

Weaknesses:

There are numerous statements in the introduction and discussion sections that I currently do not agree with and consider need to be addressed. Therefore, I recommend major revisions.

Major comments:**Introduction and Discussion:**

The introduction and the discussion are currently too general and not focused. Furthermore, there are some key concepts that are missing and are important for the reader to have an overview of the current state-of-the-art regarding endolysins that target gram-negatives. Specifically, the concepts of 'Artilysins', 'Innolysins', and 'Lysocins' are not introduced. Besides this, the authors do not mention other high-throughput mining or engineering strategies for endolysins, such as e.g. the VersaTile platform, which was initially developed by Hans Gerstmans et al. for one of the targeted pathogens in this manuscript (i.e., *Acinetobacter baumannii*). Recent works by Niels Vander Elst et al. have demonstrated that this VersaTile platform can be used to high-throughput screen and hit-to-lead select endolysins in the magnitude tens of thousands. Lastly, Roberto Vázquez et al. have recently demonstrated with bio-informatic analyses that approximately 30% of Gram-negative endolysin entries have AMP-like regions (hydrophobic short sequences), and that these entries are interesting candidates for further wet lab testing due to their outer membrane penetrating capacities. Therefore, I fully disagree with the statement being made in the introduction that endolysin strategies to target Gram-negatives are 'in its infancy' and I urge the authors to provide a new introduction that properly gives an overview of the Gram-negative endolysin field.

Results:

It should be mentioned that the halo assay is a qualitative assay for activity testing. I personally do not like that the size of the halos is used to discriminate in endolysin activity. In this reviewer's opinion, the size of the halo is highly dependent on (i) the molecular size of the endolysin as smaller proteins can diffuse further in the agar, and (ii) the affinity of the CBD subdomain of the endolysin for the bacterial peptidoglycan. It should also be said that in the halo assay, there is a long contact time between the endolysin and the bacteria that are statically embedded in the agar, which can result in false positive results. How did the authors mitigate this?

Testing should have been done at equimolar concentrations. If the authors decided to e.g. test 50 µg/mL for each protein, how was this then compensated for differences in molecular weight? For example, if PHAb10 and PHAb11 have smaller molecular sizes than PHAb7, 8, and 9, there is more protein present in 50 µg/mL for the first two compared to the others, and this would explain the higher decrease in bacterial killing (and possibly the larger halos).

<https://doi.org/10.7554/eLife.98266.1.sa1>

Reviewer #2 (Public Review):

Summary:

The study explores a new strategy of lysin-derived antimicrobial peptide-primed screening to find peptidoglycan hydrolases from bacterial proteomes. Using this strategy authors identified five peptidoglycan hydrolases from *A. baumannii*. They further tested their antimicrobial activities on various Gram-positive and Gram-negative pathogens.

Strengths:

Overall, the study is good and adds new members to the peptidoglycan hydrolases family. The authors also show that these lysins have bactericidal activities against both Gram-positive and Gram-negative bacteria. The crystal structure data is good, and reveals different thermostability to the peptidoglycan hydrolases. Structural data also reveals that PhAb10 and PHAb11 form thermostable dimers and data is corroborated by generating variant protein defective in supporting intermolecular bond pairs. The mice bacterial infection shows promise for the use of these hydrolases as antimicrobial agents.

Weaknesses:

While the authors have employed various mechanisms to justify their findings, some aspects are still unclear. Only CFU has been used to test bactericidal activity. This should also be corroborated by live/dead assay. Moreover, SEM or TEM analysis would reveal the effect of these peptidoglycan hydrolases on Gram-negative /Gram-positive cell envelopes. The authors claim that these hydrolases are similar to T4 lysozyme, but they have not correlated their findings with already published findings on T4 lysozyme. T4 lysozyme has a C-terminal amphipathic helix with antimicrobial properties. Moreover, heat, denatured lysozyme also shows enhanced bactericidal activity due to the formation of hydrophobic dimeric forms, which are inserted in the membrane. Authors also observe that heat-denatured PHAb10 and PHAb11 have bactericidal activity but no enzymatic activity. These findings should be corroborated by studying the effect of these holoenzymes/ truncated peptides on bacterial cell membranes. Also, a quantitative peptidoglycan cleavage assay should be performed in addition to the halo assay. Including these details would make the work more comprehensive.

<https://doi.org/10.7554/eLife.98266.1.sa0>